A Project Report on

ONLINE LEARNING PLATFORM WITH AI MENTORSHIP AND STARTUP MICRO-INTERNSHIPS (COURSEHUB)

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
INFORMATION TECHNOLOGY

Submitted By

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2023 – 2027

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CERTIFICATE

This is to certify that the Project report titled **"ONLINE LEARNING PLATFORM WITH AI MENTORSHIP AND STARTUP MICRO-INTERNSHIPS"** is a bonafide work done and submitted by **VADDIGALLA RISHIKANTH (HT.NO: 24EG512A04)**, in partial fulfillment of the requirements for the award of **B.Tech in Information Technology** to **Anurag University (Formerly Anurag Group of Institutions / CVSR College of Engineering)**, Hyderabad during the academic year 2023–2027 and the bonafide work has not been submitted elsewhere for the award of any other degree.

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I hereby declare that the project work entitled **"ONLINE LEARNING PLATFORM WITH AI MENTORSHIP AND STARTUP MICRO-INTERNSHIPS"**, submitted to **Anurag University** in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology in Information Technology**, is an authentic record of original work carried out by me under the guidance of **Mr. G. SEKHAR REDDY**, Assistant Professor, Department of Information Technology.

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ABSTRACT

Traditional web-based learning management systems (LMS) provide passive educational materials such as pre-recorded video lectures and downloadable PDFs, but lack economic and career conversion mechanisms for students. After acquiring practical technical skills, learners struggle to break into the industry because multinational corporations (MNCs) enforce high filtration barriers, arbitrary academic percentage thresholds (such as 90%+ criteria), multiple interview rounds, and restrictive employment bonds. Concurrently, early-stage startups and small ventures require cost-effective talent to complete modular software milestones, but lack the capital to hire high-cost senior developers.

To bridge this critical industry gap, the **Online Learning Platform with AI Mentorship and Startup Micro-Internships (CourseHub)** has been designed and implemented using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)**. The system provides a unified four-role ecosystem comprising **Students**, **Instructors**, **Startups**, and **Administrators**:

1. **Students:** Access video courses, modular curriculum tracking, interactive chapter quizzes, cryptographic QR-verified certificates, an integrated AI Mentor powered by **Google Gemini**, and a dedicated **"Skill-to-Earn"** marketplace to earn milestone stipends without corporate bonds.
2. **Instructors:** Create and manage rich multimedia courses with cloud-optimized video streaming via **Cloudinary**, author assessments with AI quiz assistance, and monitor enrollment revenue metrics.
3. **Startups:** Register via a dedicated enterprise onboarding portal capturing company name, established year, headquarters location, and employee count. Startups post paid micro-internships and mini-tasks, instantly broadcasting opportunity alerts to verified learners.
4. **Administrators:** Moderate courses, supervise user accounts, resolve platform appeals, and oversee platform analytics.

The application incorporates a secure **dual-channel JWT authentication architecture** (supporting both HTTP-only cookies and Bearer headers) that ensures cross-domain compatibility. The backend is deployed on **Render** with active health probes (`https://online-learning-platofrm.onrender.com`), while the frontend is deployed on **Vercel** with client-side Single Page Application (SPA) rewrites. Comprehensive functional, integration, and security testing demonstrates that the platform delivers high availability, low latency, robust role-based access control, and practical economic empowerment for engineering learners.

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LIST OF ABBREVIATIONS

Notation	Description
API	Application Programming Interface
CDN	Content Delivery Network
CORS	Cross-Origin Resource Sharing
CSS	Cascading Style Sheets
DOM	Document Object Model
E-R	Entity - Relationship
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
JWT	JSON Web Token
LMS	Learning Management System
MERN	MongoDB, Express.js, React.js, Node.js
MNC	Multinational Corporation
NoSQL	Not Only Structured Query Language
ODM	Object Data Modeling
PDF	Portable Document Format
QR	Quick Response Code
RBAC	Role-Based Access Control
REST	Representational State Transfer
SPA	Single Page Application
SSL	Secure Sockets Layer
TLS	Transport Layer Security
UI	User Interface
URI	Uniform Resource Identifier
UX	User Experience

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The digital revolution in tertiary education has reshaped instructional delivery worldwide. As online learning expands beyond emergency remote teaching, modern learners demand flexible, self-paced, and comprehensive education environments that empower them both academically and financially. Traditional higher education formats and static course directories provide lecture videos, but rarely connect students with immediate, practical work opportunities.

The **Online Learning Platform with AI Mentorship and Startup Micro-Internships (CourseHub)** addresses this challenge. Developed as a full-stack web application on the **MERN (MongoDB, Express.js, React.js, Node.js)** architecture, the platform unifies curriculum distribution with an experiential economic framework.

The platform provides dedicated, role-tailored capabilities for four distinct stakeholders:

- **Learners (Students):** Enroll in modular courses, stream high-definition video lectures, track course completion progress, consult a 24/7 AI Mentor powered by **Google Gemini**, earn digitally signed and QR-verified certificates, and apply for paid startup micro-internships without restrictive barriers.
- **Instructors:** Design courses, upload multimedia lectures and study resources to cloud storage, author dynamic assessments with AI-assistance, and review student performance analytics.
- **Startups & Hiring Partners:** Onboard through a dedicated enterprise portal specifying company name, established year, headquarters location, and team size. Startups publish micro-tasks and paid internships, broadcast opportunities to verified learners, and hire motivated talent with zero corporate bond friction.
- **Administrators:** Oversee platform integrity, review published courses, manage user roles, and resolve user appeals through an administrative control panel.

By pairing AI-assisted mentorship with startup project outsourcing, CourseHub establishes a sustainable educational ecosystem that reduces startup production costs while directly funding and upskilling student careers.

1.2 OBJECTIVES

The central objectives of the Online Learning Platform project are as follows:

1. **Architect a Scalable Full-Stack Web Platform:** Build a responsive, cloud-native web application using MongoDB Atlas, Express.js, React 19, and Node.js.
2. **Implement Role-Based Access Control (RBAC):** Provide granular authentication and authorization mechanisms for Students, Instructors, Startups, and Administrators using JSON Web Tokens (JWT) and dual-channel cookie/header transport.
3. **Establish a Dedicated Startup Onboarding Workflow:** Separate corporate registration from general student onboarding to capture critical organizational metadata including Company Name, Established Year, Location, and Employee Count.
4. **Engineer the "Skill-to-Earn" Micro-Internship Gateway:** Eliminate traditional 90% score gates and restrictive corporate bonds, allowing learners with verified competencies to undertake paid mini-tasks for early-stage startups.
5. **Integrate Real-Time Generative AI Mentorship:** Utilize Google Gemini Large Language Model APIs to deliver 24/7 interactive tutoring, programming guidance, and automated assessment generation.
6. **Deploy Cryptographic QR-Code Verifiable Credentials:** Issue tamper-proof course completion certificates verifiable via dynamic QR codes at `/verify-certificate`.
7. **Ensure Production Reliability & Cloud Deployment:** Deploy the backend on **Render** (`https://online-learning-platofrm.onrender.com`) and the frontend on **Vercel** with full CORS and SPA rewrite handling.

1.3 PROBLEM FORMULATION

Conventional educational platforms and university management portals suffer from fundamental structural deficiencies:

1. **Disconnection from Immediate Economic Reality:** Students spend hundreds of hours mastering programming, design, and analytics frameworks without gaining hands-on commercial experience or supplemental income.
2. **High Filtration Barriers in Corporate Recruiting:** Top-tier MNCs enforce aggressive academic score filters (requiring 90%+ marks, aptitude clearances, and long multi-stage interviews), discouraging talented learners from practical project work.
3. **High Resource Costs for Early-Stage Startups:** Small businesses and seed-stage startups frequently require small, modular tasks (e.g., landing page development, API integration, UI prototyping, data cleanup) but cannot afford full-time engineering salaries or recruitment commissions.
4. **Credential Integrity Deficits:** Standard certificates distributed as unverified PDFs are easily manipulated. Academic institutions require digital authentication to confirm bonafide completion.
5. **Static Pedagogy without Immediate Support:** When students encounter bugs or theoretical hurdles late at night, they lack responsive teaching assistance, leading to course abandonment.

1.4 SCOPE

The scope of this project encompasses the design, implementation, and cloud deployment of the full-stack web application:

- **Authentication Module:** Secure registration, dedicated startup onboarding, credential hashing, and session management.
- **Course Authoring & Management:** Video transcoding and media hosting on Cloudinary, chapter sequencing, curriculum pricing, and preview support.
- **Learning Interface:** Custom HTML5 video player, automatic chapter completion triggers, wishlist management, and enrollment databases.
- **Opportunity Portal:** Posting of paid micro-internships, mini-tasks, and full-time junior positions, application submission, resume review, and status tracking.
- **AI Mentorship Engine:** Gemini-backed conversational assistant contextualized to course curriculum.
- **Verification Engine:** Dynamic PDF generation using `jsPDF` and `html2canvas`, and QR verification routes.
- **Administrative Governance:** User status toggles (active/banned), role elevation, and course auditing.

1.5 FEASIBILITY STUDY

Before initiating development, a comprehensive feasibility study was conducted across three key dimensions:

1.5.1 Technical Feasibility

The platform utilizes the MERN stack—a battle-tested, highly supported open-source ecosystem. Node.js provides asynchronous non-blocking I/O ideal for concurrent API interactions. React.js utilizes a Virtual DOM to deliver sub-second UI transitions. MongoDB Atlas provides horizontal scalability and flexible document schemas. Multimedia files are offloaded to Cloudinary CDN, removing server disk bottlenecks. Technical feasibility is fully confirmed.

1.5.2 Operational Feasibility

The platform features an intuitive, glassmorphic UI styled with Tailwind CSS, requiring minimal technical training for students or startup founders. Automated notifications guide users through course milestones and hiring events. Operational feasibility is satisfied.

1.5.3 Economic Feasibility

The software leverages open-source libraries, free-tier cloud credits on MongoDB Atlas, Render, Vercel, and Cloudinary. No specialized hardware is required, rendering development and maintenance economically viable.

1.6 SYSTEM REQUIREMENTS

1.6.1 Software Requirements

Table 1.1 delineates the software environment utilized during development and production execution.

Table 1.1: Software Specifications

Component	Specification
Operating System	Windows 10/11, macOS Sequoia, or Ubuntu Linux 22.04 LTS
Frontend Framework	React.js 19.x, Vite 7.x, React Router DOM 7.x
Styling & Icons	Tailwind CSS 4.x, React Icons, Lucide Icons
Backend Runtime	Node.js (v20.x / v22.x LTS), Express.js 5.x
Database System	MongoDB Atlas (Cloud NoSQL Database), Mongoose ODM 9.x
Security & Auth	JSON Web Token (jsonwebtoken), bcrypt (Salt Rounds: 10), cookie-parser
Cloud Object Storage	Cloudinary Multimedia CDN SDK
Artificial Intelligence	Google Gemini Generative AI SDK (`@google/genai`)
Client HTTP Tool	Axios with custom Bearer token request/response interceptors
PDF & QR Engines	jsPDF, html2canvas, qrcode.react, qrcode
Version Control	Git, GitHub (`https://github.com/rishi1702-boop/Online_learning_platofrm`)
Cloud Hosting	Render (Backend Web Service), Vercel (Frontend Single Page App)

1.6.2 Hardware Requirements

Table 1.2 specifies the minimum and recommended hardware configurations.

Table 1.2: Hardware Specifications

Component	Minimum Specification	Recommended Specification
Central Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5/i7, AMD Ryzen 5/7, Apple Silicon M-Series
System Memory (RAM)	4 GB	8 GB to 16 GB DDR4/DDR5
Storage Capacity	20 GB Free Space	50 GB Solid State Drive (NVMe SSD)
Network Interface	Standard Broadband (2 Mbps)	High-Speed Fiber / Wi-Fi (15 Mbps+)
Display Resolution	1366 × 768 Pixels	1920 × 1080 (Full HD) or higher

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CHAPTER 2

REQUIREMENTS

2.1 REQUIREMENTS SPECIFICATION

The system requirements are categorized into functional requirements (capabilities provided by the application) and non-functional requirements (quality attributes and operational standards).

2.1.1 Functional Requirements (FR)

- **FR-01: User Registration & Role Selection:** The system must allow users to register as Students, Instructors, or Admins with first name, last name, email, and password.
- **FR-02: Dedicated Startup Registration:** The system must provide a separate onboarding flow for corporate entities collecting Company Name, Established Year (1950–Present), Headquarters Location, Employee Count range, Founder Name, Work Email, and Password.
- **FR-03: Dual Authentication Mechanism:** The system must verify credentials, issue JSON Web Tokens, and support both HTTP-only cookies and `Authorization: Bearer <token>` headers for cross-domain operation.
- **FR-04: Course Catalog & Category Discovery:** Students must be able to search, filter by category (Development, Design, Business, AI), view ratings, and inspect curriculum details.
- **FR-05: Multimedia Curriculum Delivery:** The platform must stream video lectures from Cloudinary, provide lesson navigation, and record chapter completion milestones.
- **FR-06: Interactive Assessment & Quizzing:** The platform must provide dynamic multiple-choice quizzes with timer controls, auto-grading, and score recording in the student profile.
- **FR-07: AI Mentorship Conversational Agent:** Students must be able to query an embedded Google Gemini AI tutor regarding lesson content, code bugs, and personalized study roadmaps.
- **FR-08: Cryptographic QR-Code Certificate Generation:** Upon 100% course curriculum completion, the platform must dynamically generate a downloadable PDF certificate bearing a unique verification hash and scannable QR code.
- **FR-09: Public Certificate Verification:** Anyone must be able to verify credential authenticity by scanning the certificate QR code or visiting `/verify-certificate?id=<ID>`.
- **FR-10: Startup Opportunity Creation:** Startups must be able to post micro-internships, mini-tasks, and junior jobs specifying title, description, milestone stipend, location, deadline, and zero-bond confirmation.
- **FR-11: Instant Student Broadcast Notifications:** When a startup publishes an opportunity, the system must broadcast real-time notification alerts to active learners.
- **FR-12: Job & Gig Application Engine:** Students must be able to apply directly by submitting cover letters, resume links, GitHub profiles, and portfolio URLs.
- **FR-13: Startup Candidate Reviewer:** Startups must have an evaluation dashboard to review applications, inspect portfolio links, and transition candidate status (`UNDER_REVIEW`, `SHORTLISTED`, `HIRED`, `REJECTED`).
- **FR-14: Instructor Creator Studio:** Instructors must be able to create courses, upload video chapters and PDF attachments, edit course metadata, and inspect student enrollment statistics.
- **FR-15: Administrative Oversight:** Admins must be able to view system-wide metrics, deactivate malicious users, audit courses, and resolve student appeals.

2.1.2 Non-Functional Requirements (NFR)

- **NFR-01: Performance & Response Time:** Standard API responses must resolve within 500ms under standard network conditions. Media streaming must leverage Cloudinary CDN caching.
- **NFR-02: Security & Data Privacy:** Passwords must be hashed using bcrypt with salt factor 10. Tokens must expire within 7 days. Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF) protections must be active.
- **NFR-03: Availability & Fault Tolerance:** Cloud deployment on Render and Vercel must deliver 99.5% uptime. Render active healthcheck (`/health`) must report service availability.
- **NFR-04: Scalability:** The stateless Node.js architecture must support horizontal scaling, while MongoDB Atlas handles automatic sharding and replication.

- **NFR-05: Usability & Responsiveness:** User interfaces must follow a responsive fluid grid (Tailwind CSS) rendering cleanly across mobile, tablet, laptop, and desktop viewports.

2.2 USER INTERFACES (PAGE LAYOUT & FIELDS)

The front-end user experience is partitioned into distinct page layouts:

1. **Home Landing Page (`/`):**
   - Header with dynamic navigation (Courses, Learning Paths, Login, Register, "🚀 For Startups").
   - Hero banner with platform value proposition, call-to-action buttons.
   - Featured courses grid with thumbnail, title, instructor name, rating stars, price.
   - "Why Choose Us" showcase detailing Zero-Bond Startup Internships, AI Mentorship, and QR Certificates.
   - Platform statistics counter and footer.
2. **Standard User Registration (`/register`):**
   - Left Panel: First name, last name, email, password, show/hide password toggle, strength indicator.
   - Role selection buttons: Learn (Student), Teach (Instructor), Admin.
   - Prominent Startup Banner: "Registering as a Startup? Startups have a separate registration with company details [Startup Portal →]".
   - Submit button ("Create free account") and login redirect link.
   - Right Panel: Adaptive branding visual with benefit checkmarks.
3. **Dedicated Startup Registration Page (`/register-startup`):**
   - Left Panel: Value proposition highlighting zero-bond hiring, affordable production cost, and broadcast alerts.
   - Right Panel Form Fields:
     - *Company / Startup Name* (`companyName` - text field)
     - *Founder / Lead First Name* (`firstName` - text field)
     - *Founder Last Name* (`lastName` - text field)
     - *Official Work Email* (`email` - email input with regex validation)
     - *Password* (`password` - password input with eye icon toggle)
     - *Established Year* (`establishedYear` - number input, range 1950 to present)
     - *Number of Employees* (`noOfEmployees` - dropdown: 1-10 Early Stage, 11-50 Seed/Growth, 51-200 Expansion, 200+ Enterprise)
     - *Headquarters / Location* (`location` - text field, e.g., Bengaluru, India or Remote)
     - *Company Website* (`website` - optional URL field)
   - Submit button ("Register Startup Account") and "Sign In to Startup Portal" link.
4. **User Authentication Page (`/login`):**
   - Form Fields: Email Address, Password, Forgot Password link.
   - Submit button ("Sign In" with spinning loader).
   - Helper links: "Don't have an account? Create an account" and "Hiring as a startup? Register your Startup →".
   - Client routing to `/student-dashboard`, `/instructor-dashboard`, `/startup-dashboard`, or `/admin-dashboard`.
5. **Student Dashboard (`/student-dashboard`):**
   - Welcome banner with learner name and active streak badge.
   - Stat cards: Courses in Progress, Completed Courses, Certificates Earned, Skill Points.
   - Enrolled courses carousel with progress bars ("Continue Learning" button).
   - Recommended micro-internships feed.
6. **Course Content & Video Player (`/learn/:courseId`):**

- Left/Main Viewport: Custom video player streaming Cloudinary MP4, chapter title, study notes tab, AI mentor tab.
- Right Sidebar: Modular accordion of curriculum modules and video lessons. Checkmarks indicate completed lessons.
- "Claim Certificate" button activates upon 100% completion.

7. **Startup Internships & Mini-Tasks Portal (`/jobs`):**
   - Search bar and filter tabs: All Opportunities, Micro-Internships, Mini-Tasks, Full-Time Jobs.
   - Opportunity cards displaying: Title, Startup Name, Location, Stipend/Salary badge, "Zero Corporate Bond" badge, "Open to All Learners" badge.
   - "View Details & Apply" button navigating to `/jobs/:id`.
8. **Opportunity Details & Application Page (`/jobs/:id`):**
   - Header with company profile, founded year, location, employee count.
   - Detailed project description, deliverables, required skills, stipend milestone breakdown.
   - Modal Application Form: Full Name, Email, Resume link (Google Drive / Cloudinary), GitHub URL, Portfolio URL, Cover Letter.
   - Submit button with instant validation and confirmation toast.
9. **Startup Management Dashboard (`/startup-dashboard`):**
   - Top metrics: Active Postings, Total Applicants, Hired Candidates, Open Budget.
   - Quick Action: "+ Post New Internship / Mini-Task" button.
   - Tab 1: "My Active Postings" table with edit, close, and applicant count buttons.
   - Tab 2: "Applicant Reviewer" table displaying applicant name, applied role, resume icon, portfolio link, date, and status selector (`UNDER_REVIEW`, `SHORTLISTED`, `HIRED`, `REJECTED`).
10. **Post Internship / Job Page (`/startup/post-internship`):**
    - Form Fields: Opportunity Title, Type selector (Mini-Task, Internship, Full-Time), Category, Stipend amount & schedule, Project Deliverables, Required Skills tags, Application Deadline.
    - Checkbox: "Zero Bond Confirmation (Guarantee no forced bonds for students)".
    - Broadcast toggle: "Broadcast notification to eligible learners upon posting".
    - Publish button.
11. **Certificate Verification Page (`/verify-certificate`):**
    - Search input: Certificate ID.
    - Verified Certificate Card: Learner Name, Course Title, Issuing Authority ("CourseHub Verified"), Completion Date, Digital Hash, and QR confirmation.

2.3 CONSTRAINTS & PREREQUISITES

2.3.1 Security Constraints

- Passwords must be at least 6 characters in length and include uppercase and numerical elements.
- Sensitive endpoints must strictly validate JWT signatures using secret environment variables (`JWT_SECRET`).
- Third-party cookies must be handled gracefully: since modern browsers restrict cross-domain cookies between Vercel and Render, the system must utilize Authorization Bearer headers as a persistent fallback.

2.3.2 Operational Constraints

- Media assets must not exceed 100 MB per video upload.
- File uploads are streamed directly to Cloudinary to preserve server memory.

2.3.3 Prerequisites

- Node.js environment >= 18.0.0 and <= 22.x LTS.

- Active internet connectivity to establish database handshakes with MongoDB Atlas cloud clusters.
- Valid API credentials for Cloudinary and Google Gemini.

CHAPTER 3

ANALYSIS

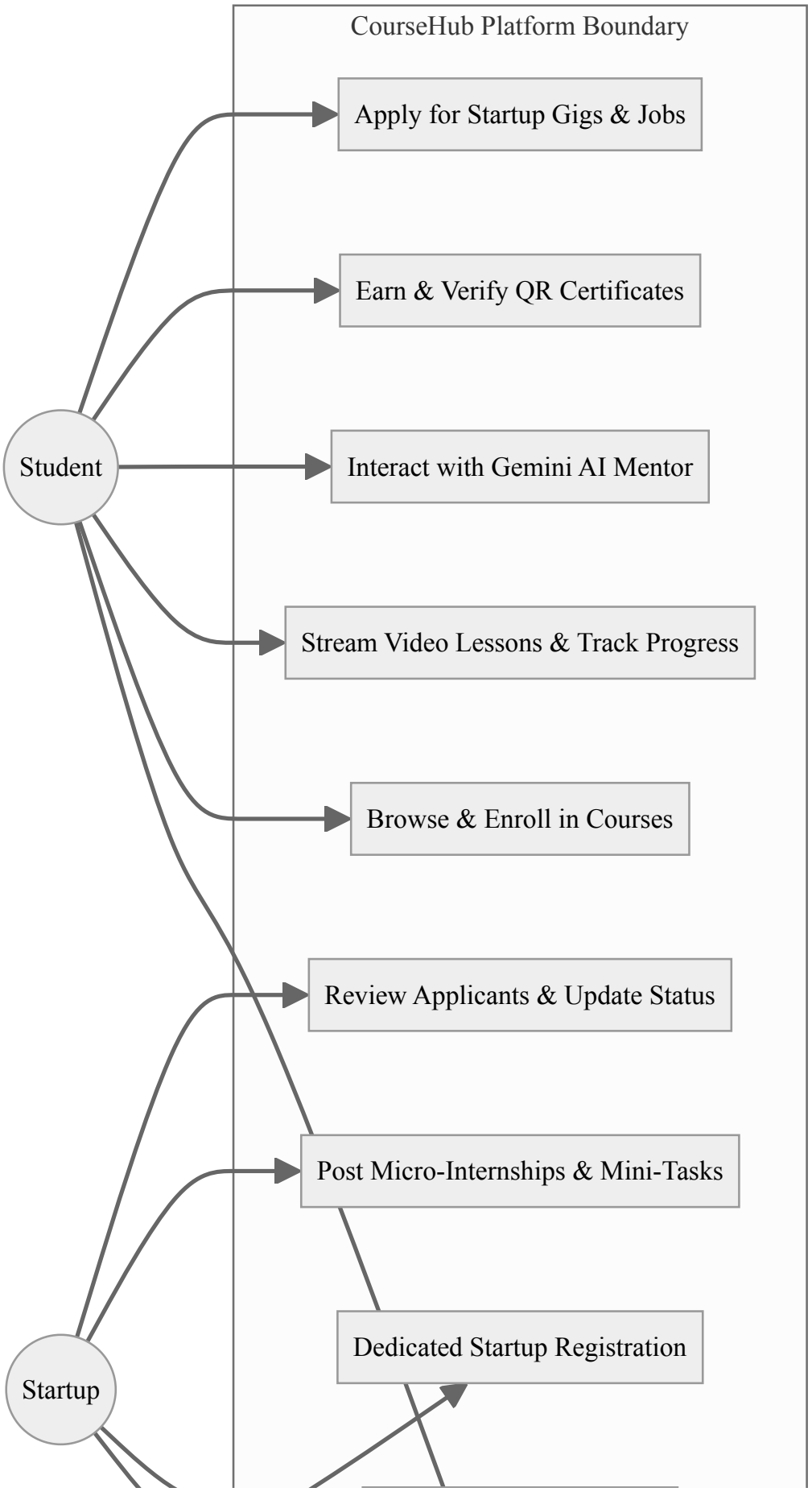
3.1 USE CASE MODEL

The use case model captures the interactions between external actors and system capabilities.

3.1.1 Use Case Diagram

The primary actors interacting with CourseHub are:

1. **Student:** Enrolls in courses, watches videos, queries AI mentor, applies for gigs, downloads certificates.
2. **Instructor:** Creates curriculum, uploads lectures, builds quizzes, views revenue analytics.
3. **Startup Partner:** Registers company, publishes gigs/jobs, reviews student portfolios, updates hiring status.
4. **Administrator:** Manages user accounts, audits courses, resolves platform appeals.


CourseHub Platform Boundary
Apply for Startup Gigs & Jobs
Earn & Verify QR Certificates
Student
Interact with Gemini AI Mentor
Stream Video Lessons & Track Progress
Browse & Enroll in Courses
Review Applicants & Update Status
Post Micro-Internships & Mini-Tasks
Dedicated Startup Registration
Startup

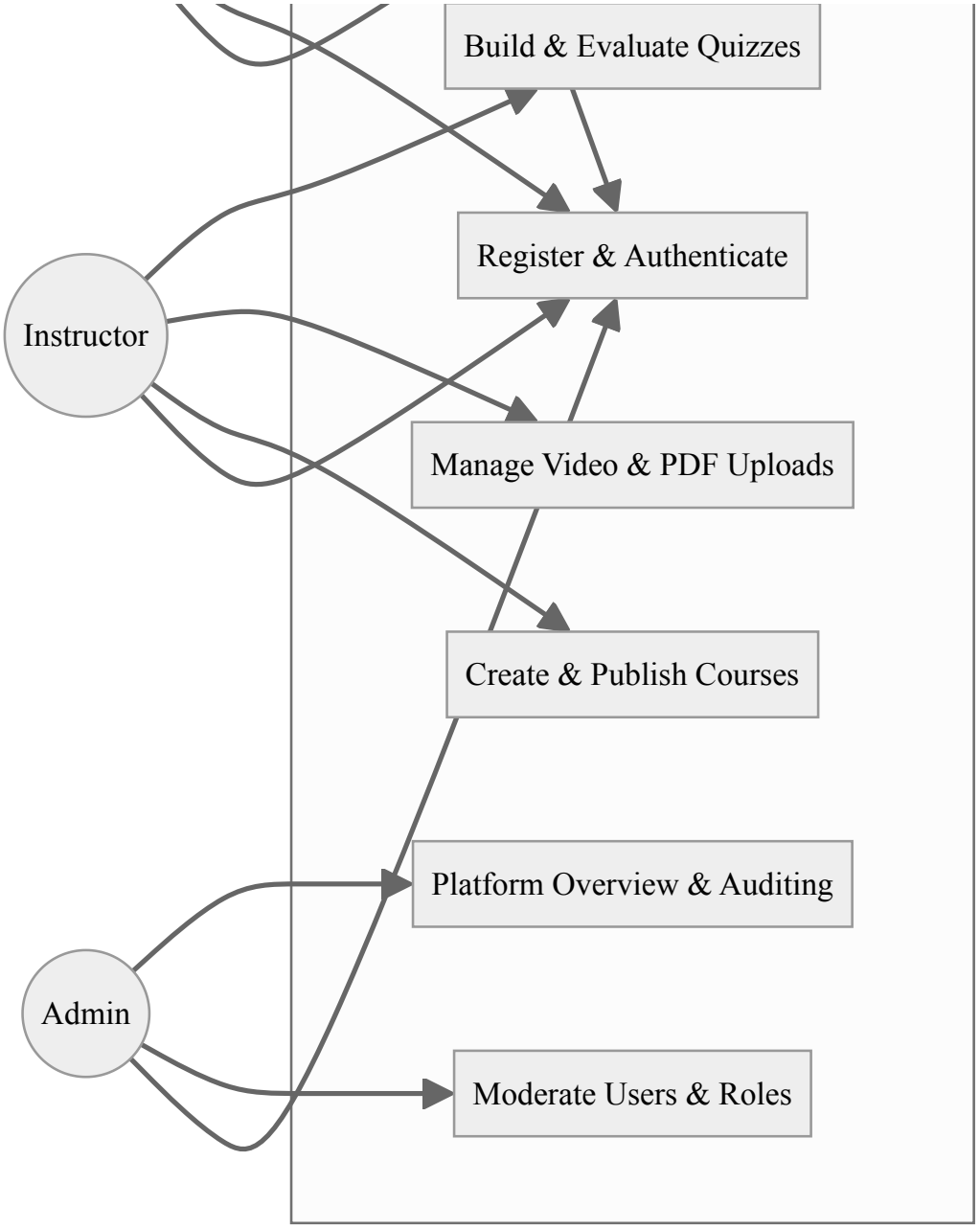


Figure 3.1: Platform Overall Use Case Diagram

3.1.1.1 Flow of Events (for each use case)

Use Case UC-01: Dedicated Startup Registration

- **Actor:** Startup Founder / Corporate Recruiter.
- **Preconditions:** The recruiter possesses an official work email and corporate metadata.
- **Main Flow:**
  1. Recruiter navigates to `/register-startup`.
  2. Enters Company Name, Founder Name, Work Email, Password, Established Year, Employee Count, and Location.
  3. Clicks "Register Startup Account".
  4. Backend checks if email is unique, hashes password via bcrypt, initializes `startupProfile`, and stores record with `role: "STARTUP"`.
  5. System presents success toast and routes user to `/login`.

- **Alternative Flow:**
  - *Email already exists:* System displays error "An account with this email already exists" and keeps form data intact.
- **Postconditions:** A new startup profile exists in MongoDB Atlas, ready to post opportunities.

Use Case UC-02: Course Enrollment & Progress Tracking

- **Actor:** Student.
- **Preconditions:** Student is logged in with active credentials.
- **Main Flow:**
  1. Student browses `/all-courses` and clicks a desired course card.
  2. Clicks "Enroll Now" on `/course/:courseId`.
  3. Course ID is added to `coursesEnrolled` array in Student's user document.
  4. Student opens course player `/learn/:courseId` and watches video lectures.
  5. As each video completes, frontend dispatches lesson completion request updating progress percentage.
- **Postconditions:** Student progress reaches 100%, unlocking certificate generation.

Use Case UC-03: Startup Gig Posting & Instant Broadcast

- **Actor:** Startup Partner.
- **Preconditions:** Authenticated as `STARTUP` role.
- **Main Flow:**
  1. Startup opens `/startup/post-internship`.
  2. Enters Opportunity Title, Type (Mini-Task/Internship), Description, Deliverables, Stipend, and Deadline.
  3. Checks "Zero Bond Guarantee" and clicks "Publish Opportunity".
  4. Backend creates new `Gig` document linked to startup's `_id`.
  5. Backend generates notification records in `Notification` collection for active students.
- **Postconditions:** The micro-internship appears on the student `/jobs` board, and notifications appear in student header dropdowns.

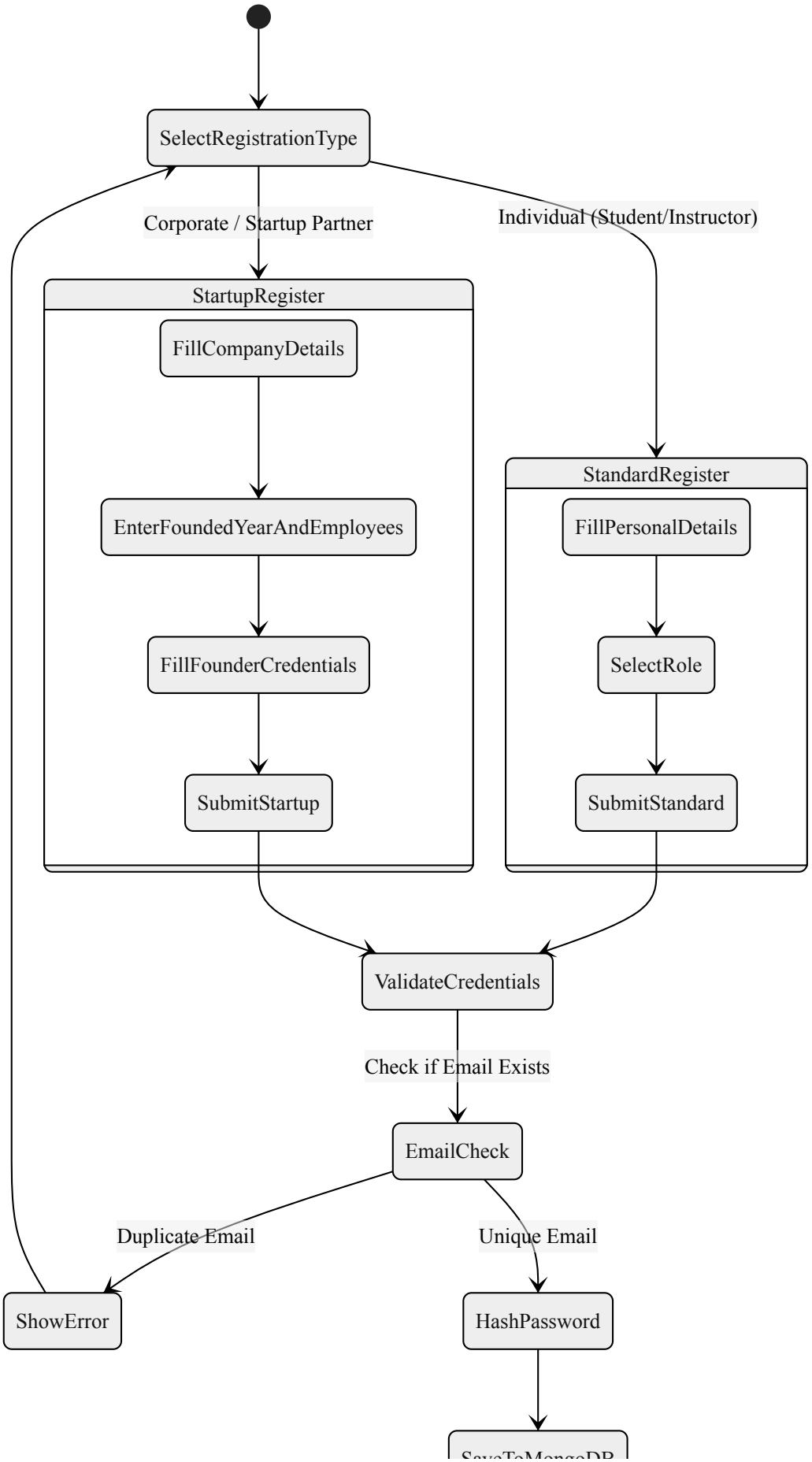
Use Case UC-04: Student Gig Application & Startup Review

- **Actor:** Student & Startup Partner.
- **Preconditions:** Opportunity is published; Student is authenticated.
- **Main Flow:**
  1. Student navigates to `/jobs/:id` and reviews deliverables and stipend.
  2. Clicks "Apply Now", attaches Resume URL, GitHub portfolio link, and cover letter, then submits.
  3. Application is created in database with status `APPLIED`.
  4. Startup accesses `/startup-dashboard` and views the applicant in "Applicant Reviewer".
  5. Startup clicks candidate resume and portfolio, assesses suitability, and changes status dropdown to `SHORTLISTED` or `HIRED`.
  6. Backend updates application status and sends notification alert to student.
- **Postconditions:** Candidate is hired without recruitment commissions or employment bonds.

3.1.2 Activity Diagrams

Activity diagrams model the step-by-step procedural workflow of operations within the platform.

Activity 1: User & Startup Registration Workflow


SelectRegistrationType
Corporate / Startup Partner
Individual (Student/Instructor)
StartupRegister
FillCompanyDetails
EnterFoundedYearAndEmployees
FillFounderCredentials
SubmitStartup
StandardRegister
FillPersonalDetails
SelectRole
SubmitStandard
ValidateCredentials
Check if Email Exists
EmailCheck
Duplicate Email
Unique Email
ShowError
HashPassword

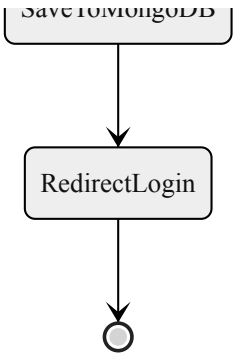
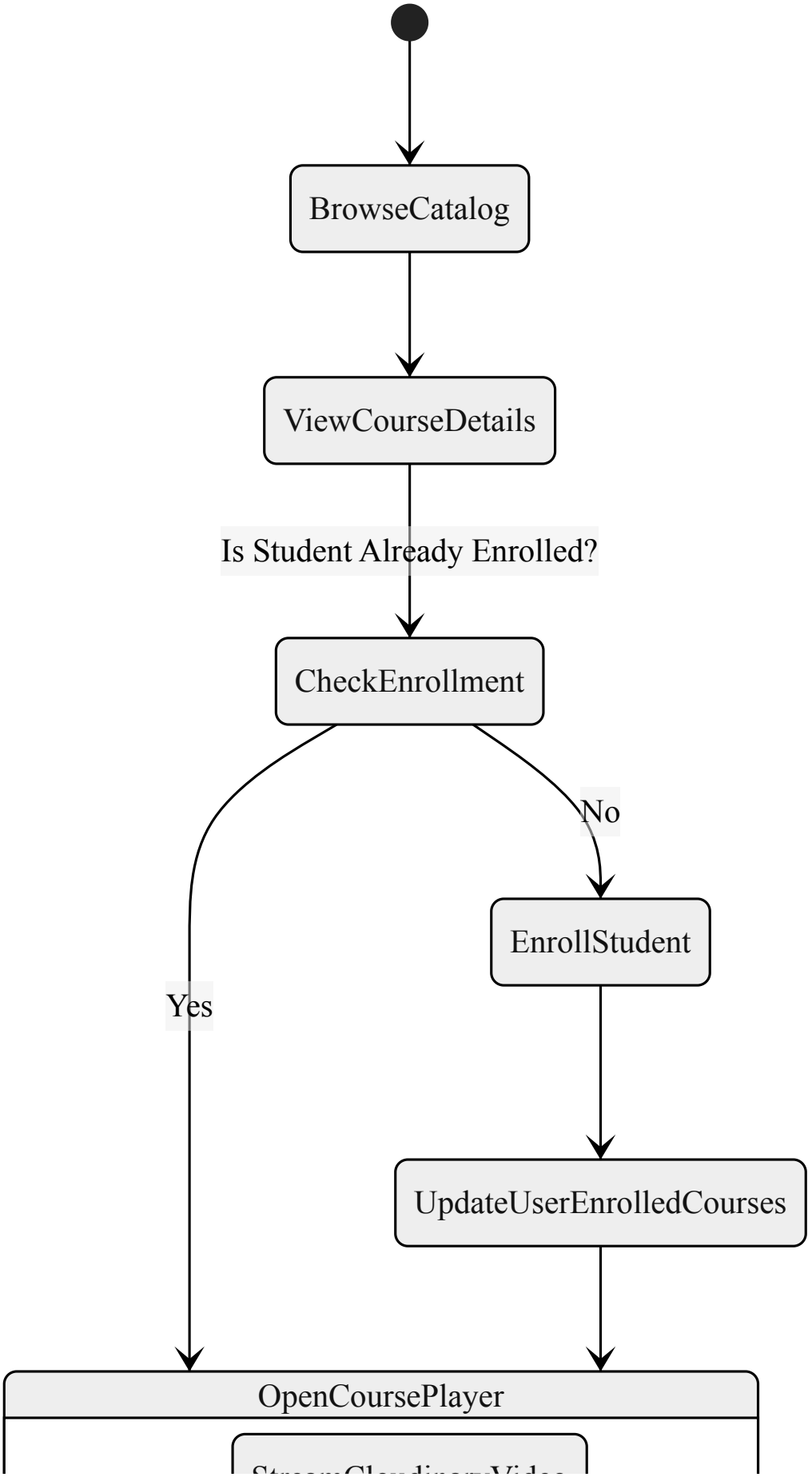


Figure 3.2: Activity Diagram for User & Startup Registration

Activity 2: Course Enrollment & Learning Progress Workflow


BrowseCatalog
ViewCourseDetails
Is Student Already Enrolled?
CheckEnrollment
No
Yes
EnrollStudent
UpdateUserEnrolledCourses
OpenCoursePlayer

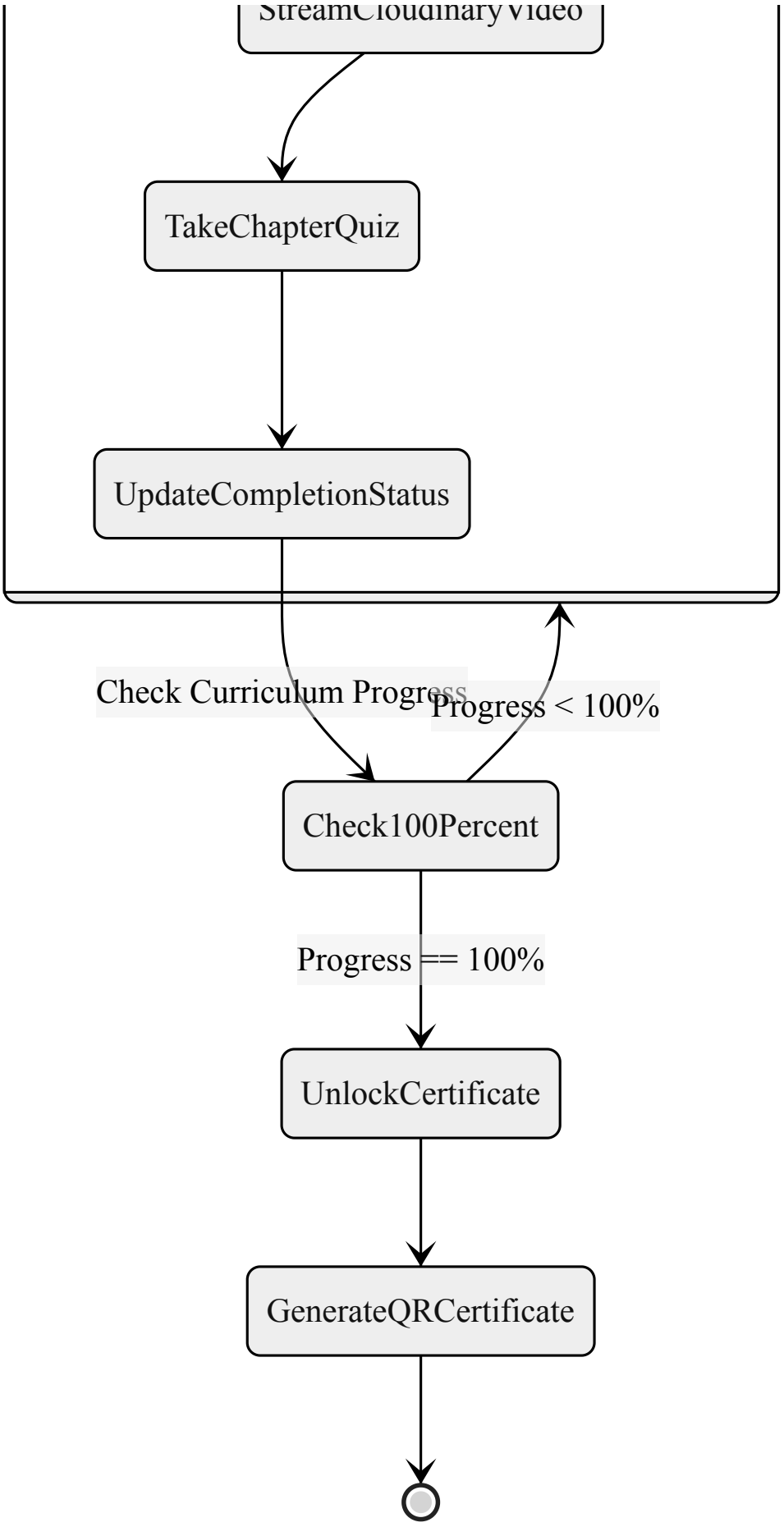
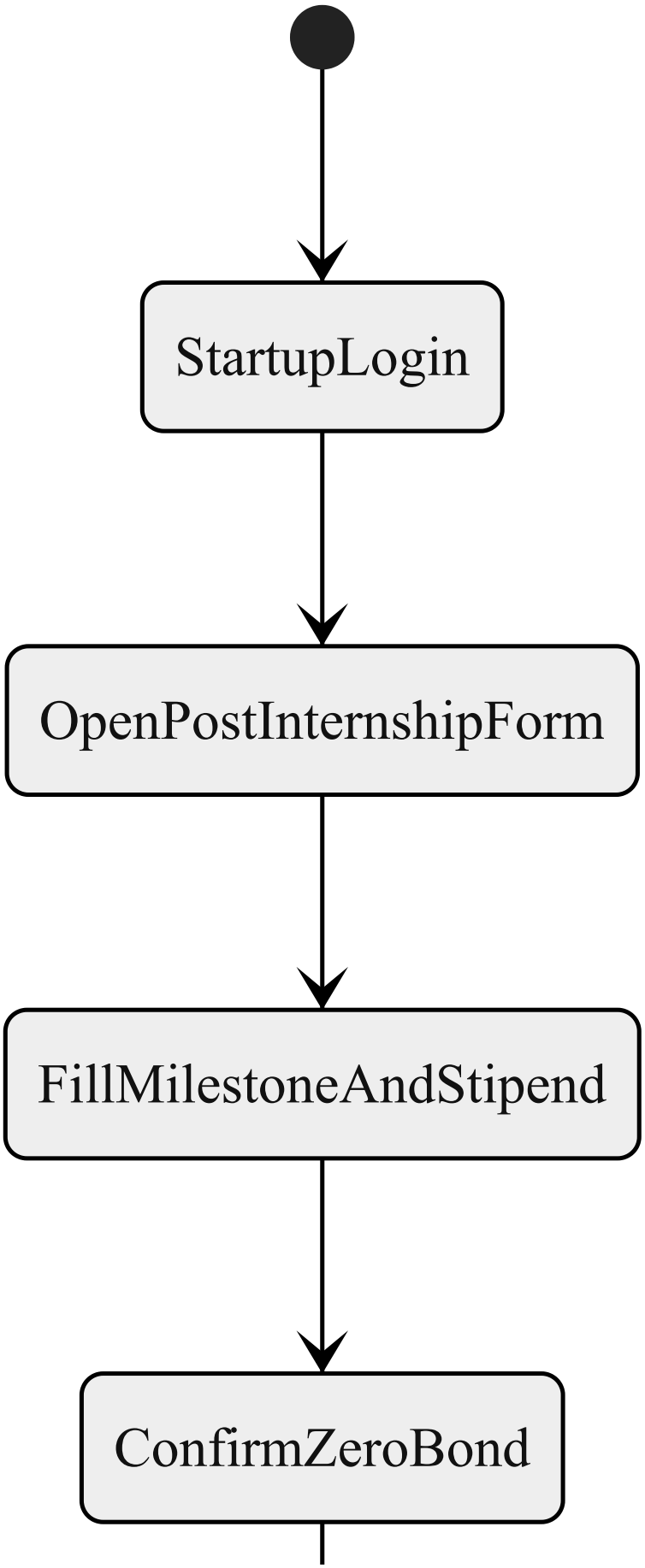
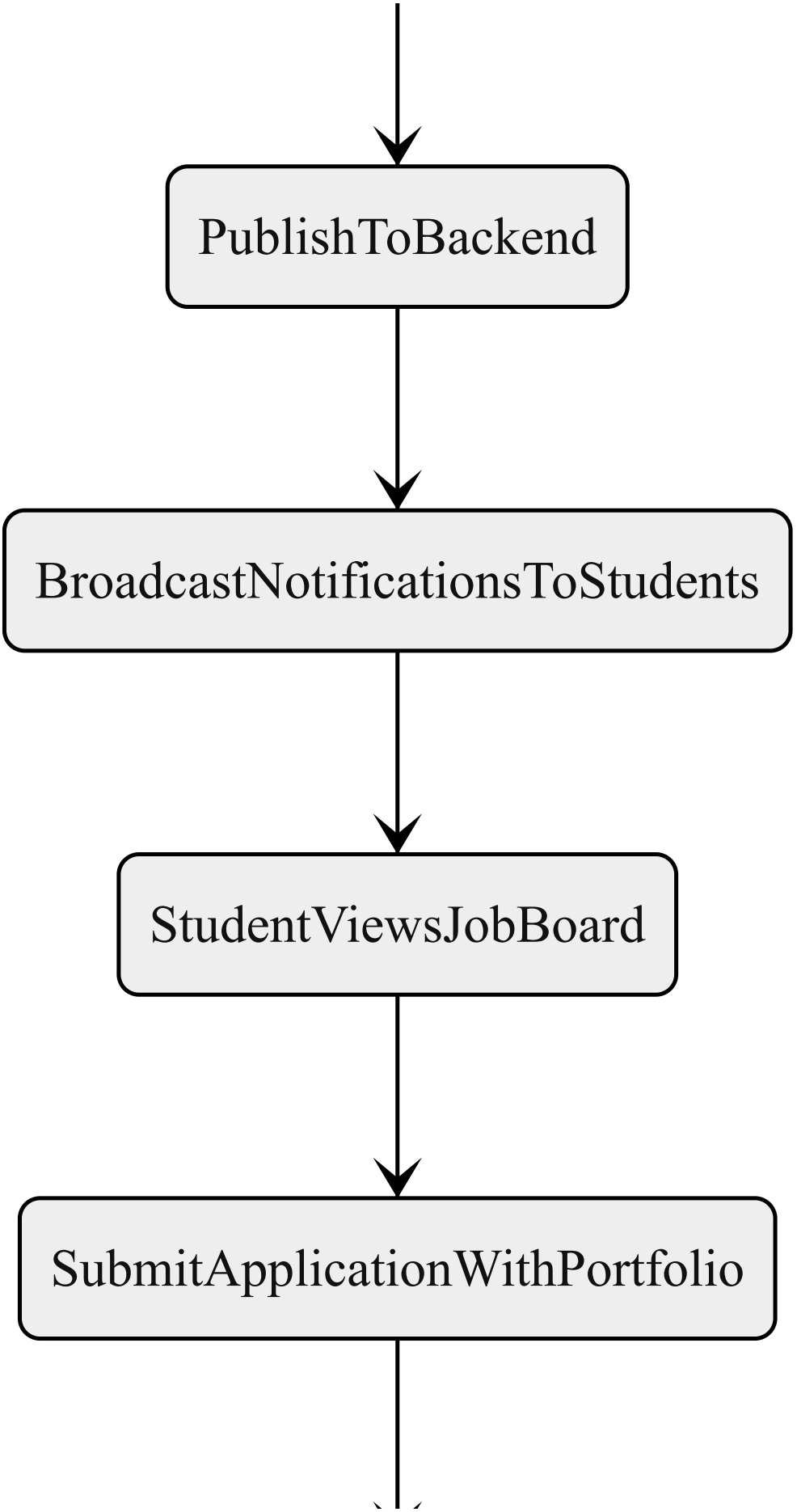

StreamCloudinaryVideo
TakeChapterQuiz
UpdateCompletionStatus
Check Curriculum Progress
Progress < 100%
Check100Percent
Progress == 100%
UnlockCertificate
GenerateQRCertificate

Figure 3.3: Activity Diagram for Course Enrollment & Learning Progress

Activity 3: Startup Gig Posting & Application Flow


StartupLogin
OpenPostInternshipForm
FillMilestoneAndStipend
ConfirmZeroBond


PublishToBackend
BroadcastNotificationsToStudents
StudentViewsJobBoard
SubmitApplicationWithPortfolio

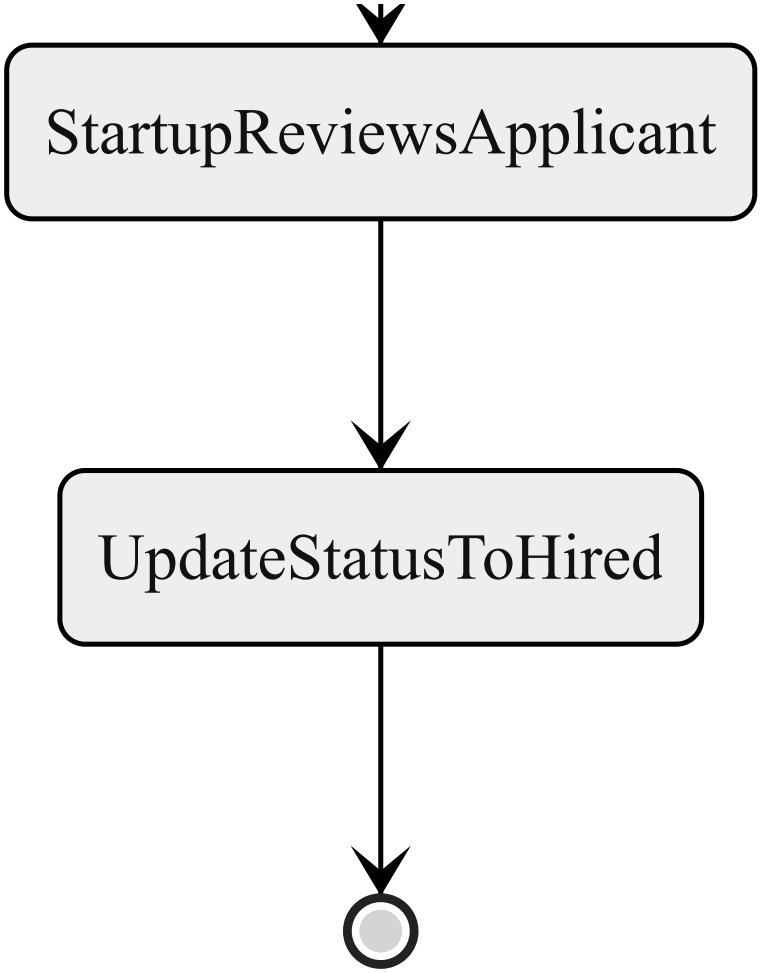


Figure 3.4: Activity Diagram for Startup Gig Posting & Application Flow

CHAPTER 4

DESIGN

4.1 DATABASE DESIGN

The data layer uses **MongoDB Atlas**, an enterprise cloud NoSQL database, managed via the **Mongoose ODM**. Below are the data schemas:

Table 4.1: User Collection Schema (`users`)

Field Name	Data Type	Constraints	Description
`_id`	ObjectId	Primary Key, Auto	Unique document identifier
`firstName`	String	Required, Trim	User's first name
`lastName`	String	Required, Trim	User's last name
`email`	String	Required, Unique, Lowercase	User's registered email
`password`	String	Required, Min: 6	Bcrypt hashed password string
`role`	String	Enum: STUDENT, INSTRUCTOR, STARTUP, ADMIN	Role-based permission tier
`startupProfile.companyName`	String	Optional (Required if STARTUP)	Registered company name
`startupProfile.establishedYear`	String	Optional	Year company was established
`startupProfile.location`	String	Default: 'Bengaluru, India'	Corporate headquarters location
`startupProfile.noOfEmployees`	String	Default: '1-10'	Team size range (e.g., 1-10, 11-50)
`startupProfile.website`	String	Optional	Company homepage URL
`coursesEnrolled`	Array of ObjectId	Ref: 'Course'	List of enrolled course IDs
`wishlist`	Array of ObjectId	Ref: 'Course'	Saved course wishlist IDs
`learningStreak.currentStreak`	Number	Default: 0	Consecutive days active
`isActive`	Boolean	Default: true	Account moderation status
`createdAt` / `updatedAt`	Date	Timestamps	Audit timestamps

Table 4.2: Course Collection Schema (`courses`)

Field Name	Data Type	Constraints	Description
`_id`	ObjectId	Primary Key	Course identifier
`title`	String	Required, Trim	Descriptive course title
`description`	String	Required	Full curriculum overview
`category`	String	Required	Category classification

Field Name	Data Type	Constraints	Description
price	Number	Required, Min: 0	Course enrollment fee
thumbnail	String	Required	Cloudinary image asset URL
instructor	ObjectId	Required, Ref: 'User'	Creator instructor ID
modules	Array of Objects	Structured	Sections containing video lectures
isPublished	Boolean	Default: true	Visibility status
averageRating	Number	Default: 5.0	Aggregated student rating

Table 4.3: Gig / Opportunity Collection Schema (`gigs`)

Field Name	Data Type	Constraints	Description
_id	ObjectId	Primary Key	Opportunity identifier
title	String	Required	Internship / Mini-task title
companyName	String	Required	Posting startup name
opportunityType	String	Enum: MINI_TASK, INTERNSHIP, JOB	Classification of work
stipend	String	Required	Milestone stipend amount
location	String	Required	Work location / Remote
deadline	Date	Required	Application cutoff date
hasBond	Boolean	Default: false	Strict zero-bond declaration
minimumScoreRequired	Number	Default: 0	Score threshold (set to 0 for access)
postedBy	ObjectId	Required, Ref: 'User'	Creator startup user ID

Table 4.4: Application Collection Schema (`applications`)

Field Name	Data Type	Constraints	Description
_id	ObjectId	Primary Key	Application identifier
opportunityId	ObjectId	Required, Ref: 'Gig'	Associated gig / internship
studentId	ObjectId	Required, Ref: 'User'	Applicant student ID
resumeUrl	String	Required	Link to student resume
githubUrl	String	Optional	Student GitHub portfolio link
portfolioUrl	String	Optional	Student live project portfolio
status	String	Enum: APPLIED, UNDER_REVIEW, SHORTLISTED, HIRED, REJECTED	Current candidate hiring status
coverLetter	String	Optional	Statement of interest

Table 4.5: Certificate Collection Schema (`certificates`)

Field Name	Data Type	Constraints	Description
`_id`	ObjectId	Primary Key	Certificate identifier
`studentId`	ObjectId	Required, Ref: 'User'	Recipient student ID
`courseId`	ObjectId	Required, Ref: 'Course'	Completed course ID
`certificateNumber`	String	Required, Unique	Cryptographic validation hash
`issueDate`	Date	Default: Date.now	Issue date timestamp
`verificationUrl`	String	Required	Public verification link

4.2 E - R DIAGRAM

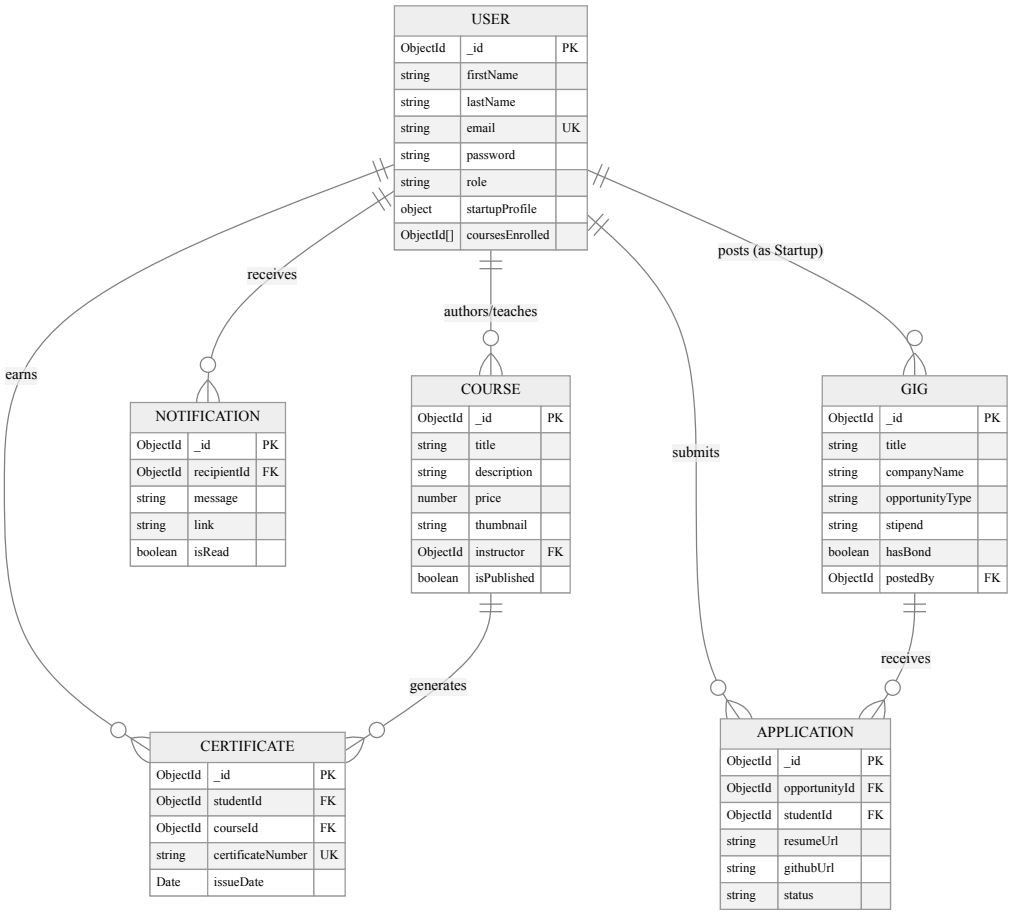


Figure 4.1: Entity-Relationship (E-R) Diagram

4.3 SEQUENCE DIAGRAMS

4.3.1 Startup Hiring & Application Lifecycle

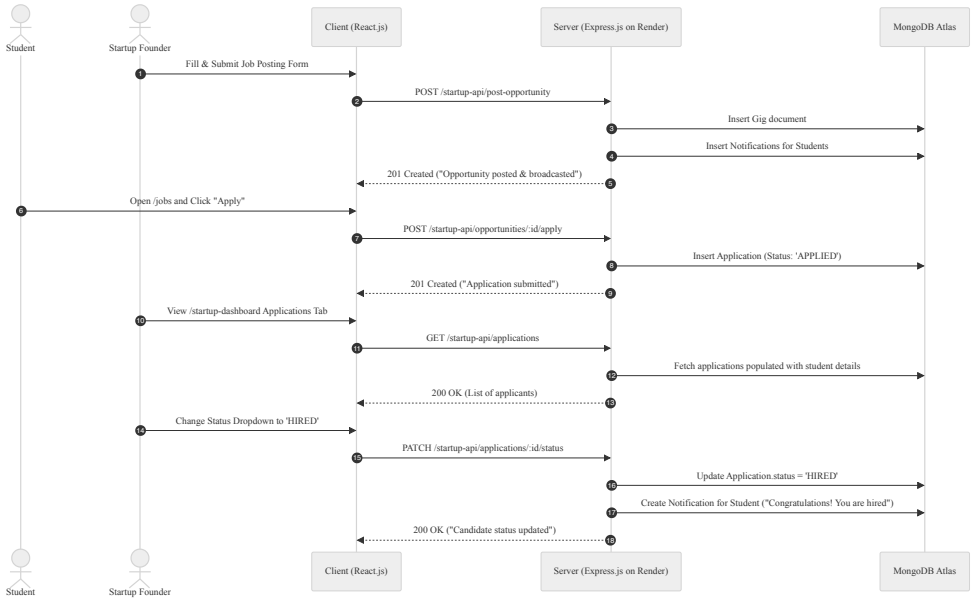


Figure 4.2: Sequence Diagram for Startup Hiring & Application Lifecycle

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4.3.2 Course Completion & Certificate Generation Flow

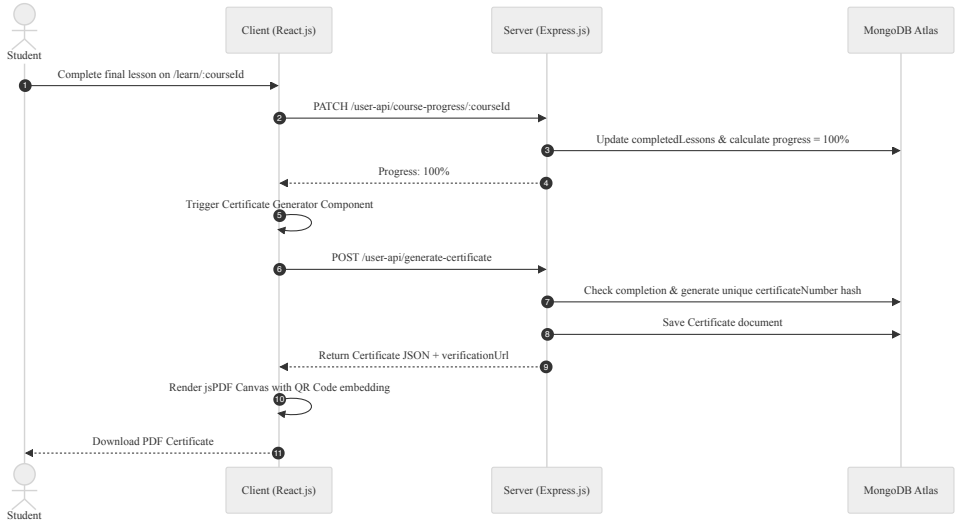


Figure 4.3: Sequence Diagram for Course Completion & Certificate Generation

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4.4 CLASS DIAGRAM

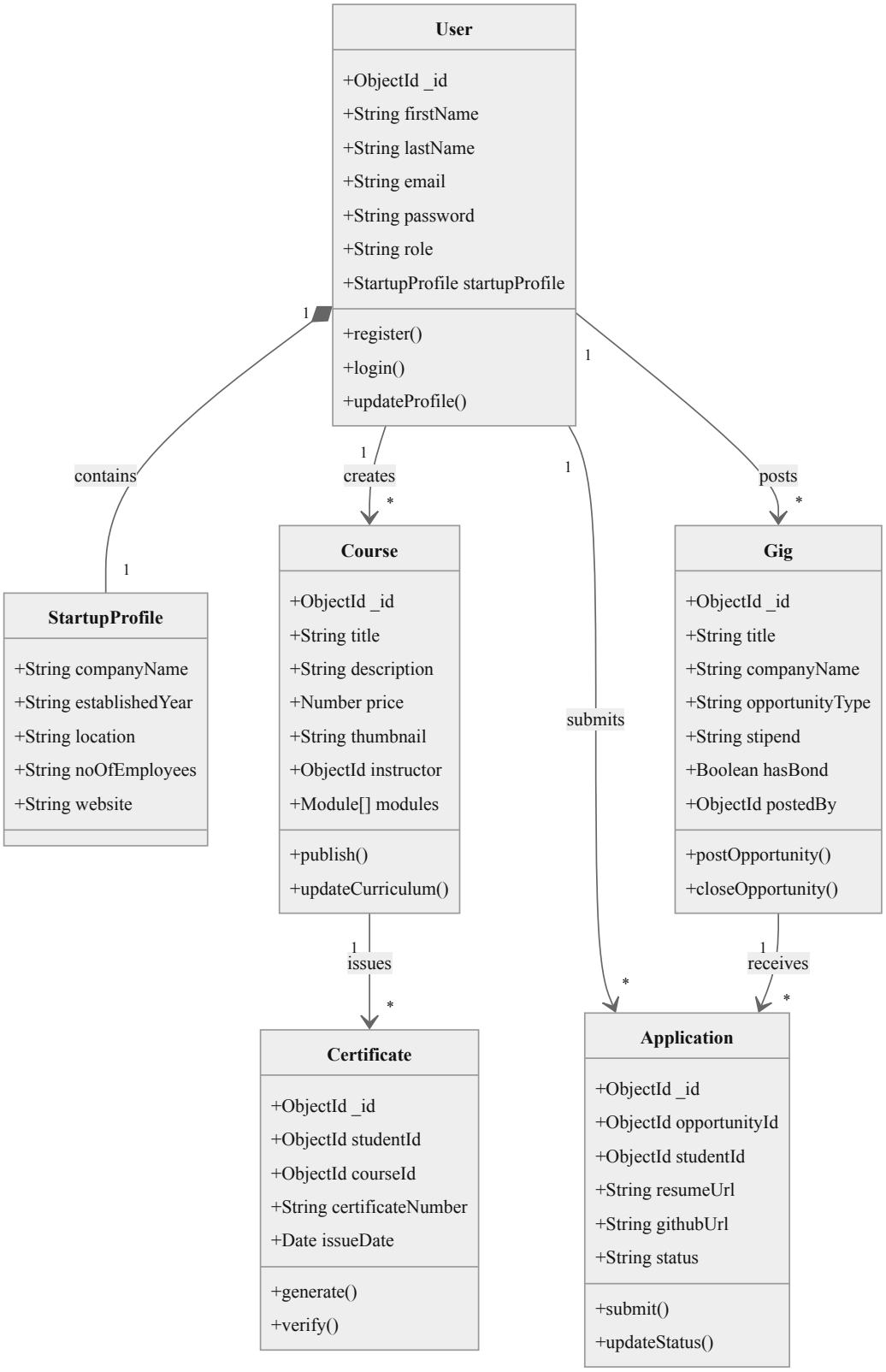


Figure 4.4: Object-Oriented Class Diagram

CHAPTER 5

IMPLEMENTATION

5.1 ARCHITECTURE & TECH STACK INTEGRATION

CourseHub is deployed using a decoupled, cloud-native three-tier topology:

```
[ User Browser / Client ]
        │
        ▼ (HTTPS)
[ Vercel CDN Edge Network ] ──> Serves React 19 SPA (Vite Production Bundle)
        │
        ▼ (REST API / JSON with Bearer Token & Cookies)
[ Render Cloud Web Service ] ──> Express.js 5 / Node.js Engine
        │
        ├──────> [ MongoDB Atlas ] (Persistent Data Storage & Sharding)
        ├──────> [ Cloudinary CDN ] (Video Lecture Streaming & Image Uploads)
        └──────> [ Google Gemini API ] (Generative AI Tutoring & Assessment Engine)
```

Figure 5.1: Three-Tier Physical Deployment Architecture

5.2 BACKEND API IMPLEMENTATION

The backend follows an Express router pattern. Key controller implementations are outlined below.

5.2.1 Dedicated Startup Registration (`backend/APIs/commonAPI.js`)

```
authRouter.post('/register-startup', async (req, res) => {
    try {
        const {
            firstName, lastName, email, password,
            companyName, establishedYear, location,
            noOfEmployees, website, about
        } = req.body;

        if (!email || !password || !companyName) {
            return res.status(400).json({ message: "Email, password, and company name are
required" });
        }

        const existingUser = await User.findOne({ email });
        if (existingUser) {
            return res.status(400).json({ message: "An account with this email already exists"
});
        }

        const salt = await bcrypt.genSalt(10);
        const hashedPassword = await bcrypt.hash(password, salt);

        const newStartup = await User.create({
            firstName: firstName || companyName,
            lastName: lastName || "Team",
            email,
            password: hashedPassword,
            role: "STARTUP",
            startupProfile: {
                companyName,
                startupName: companyName,
                establishedYear: establishedYear || new Date().getFullYear().toString(),
                location: location || "Bengaluru, India",
                noOfEmployees: noOfEmployees || "1-10",
                website: website || "",
                stage: "Early Stage",
                about: about || ""
            }
        });

        const userObj = newStartup.toObject();
        delete userObj.password;

        res.status(201).json({
            message: "Startup registered successfully! You can now log in and post
opportunities.",
            payload: userObj
        });
    } catch (error) {
        res.status(500).json({ message: error.message });
    }
});
```

5.2.2 Dual-Channel Token Verification (`backend/middlewares/verifyToken.js`)

```
export const verifyToken = (req, res, next) => {
    // Check if token exists in cookies or Authorization header
    const authHeader = req.headers.authorization;
    const bearerToken = authHeader && authHeader.startsWith('Bearer ') ? authHeader.split(' ')
[1] : null;
    const token = req.cookies?.token || bearerToken;

    if (!token) {
        return res.status(401).json({ message: "Access denied. No token provided." });
    }

    try {
        const secret = process.env.JWT_SECRET || "fallback_secret_key";
        const decoded = jwt.verify(token, secret);
        req.user = decoded;
        next();
    } catch (error) {
        res.status(401).json({ message: "Invalid or expired token." });
    }
};
```

5.3 FRONTEND COMPONENT ARCHITECTURE

5.3.1 Persistent Client Interceptor (`frontend/src/api/axiosInstance.js`)

```
export const axiosInstance = axios.create({
  baseURL: BASE_URL,
  withCredentials: true,
  timeout: 15000
});

// Attach JWT token from localStorage if present for reliable cross-domain requests
axiosInstance.interceptors.request.use(
  (config) => {
    const token = localStorage.getItem("token");
    if (token) {
      config.headers.Authorization = `Bearer ${token}`;
    }
    return config;
  },
  (error) => Promise.reject(error)
);

axiosInstance.interceptors.response.use(
  (response) => response,
  (error) => {
    if (error.response && error.response.status === 401) {
      localStorage.removeItem("user");
      localStorage.removeItem("token");
      if (window.location.pathname !== "/login") {
         window.location.href = "/login";
      }
    }
    return Promise.reject(error);
  }
);
```

5.4 CLOUD DEPLOYMENT PIPELINE (RENDER & VERCEL)

5.4.1 Render Blueprint (`render.yaml`)

```
services:
  - type: web
    name: online-learning-platform-backend
    runtime: node
    plan: free
    rootDir: backend
    buildCommand: npm install
    startCommand: node server.js
    healthCheckPath: /health
    envVars:
      - key: NODE_ENV
        value: production
      - key: PORT
        value: 10000
      - key: MONGO_URL
        sync: false
      - key: JWT_SECRET
        sync: false
```

5.4.2 Vercel Single Page App Configuration (`frontend/vercel.json`)

```
{
  "rewrites": [
    {
      "source": "/(.*)",
      "destination": "/index.html"
    }
  ]
}
```

CHAPTER 6

USER SCREENS

This chapter showcases the actual graphical user interfaces (GUI) of the deployed CourseHub application, captured directly from the live execution environment. Each figure is accompanied by an architectural breakdown of the layout regions, input components, data validation rules, and role-based permissions.

6.1 Screen 1: Home Landing Page

The primary entry point welcomes prospective learners, instructors, and corporate partners. It incorporates responsive navigation with quick links to Courses, Learning Paths, Login, Register, and a highlighted **"🚀 For Startups"** partner button. The hero section articulates the central platform mission: *"Skills that move your career forward"*, linking theoretical mastery directly to real-world career paths. The bottom section showcases corporate trust badges and a floating AI chatbot button.

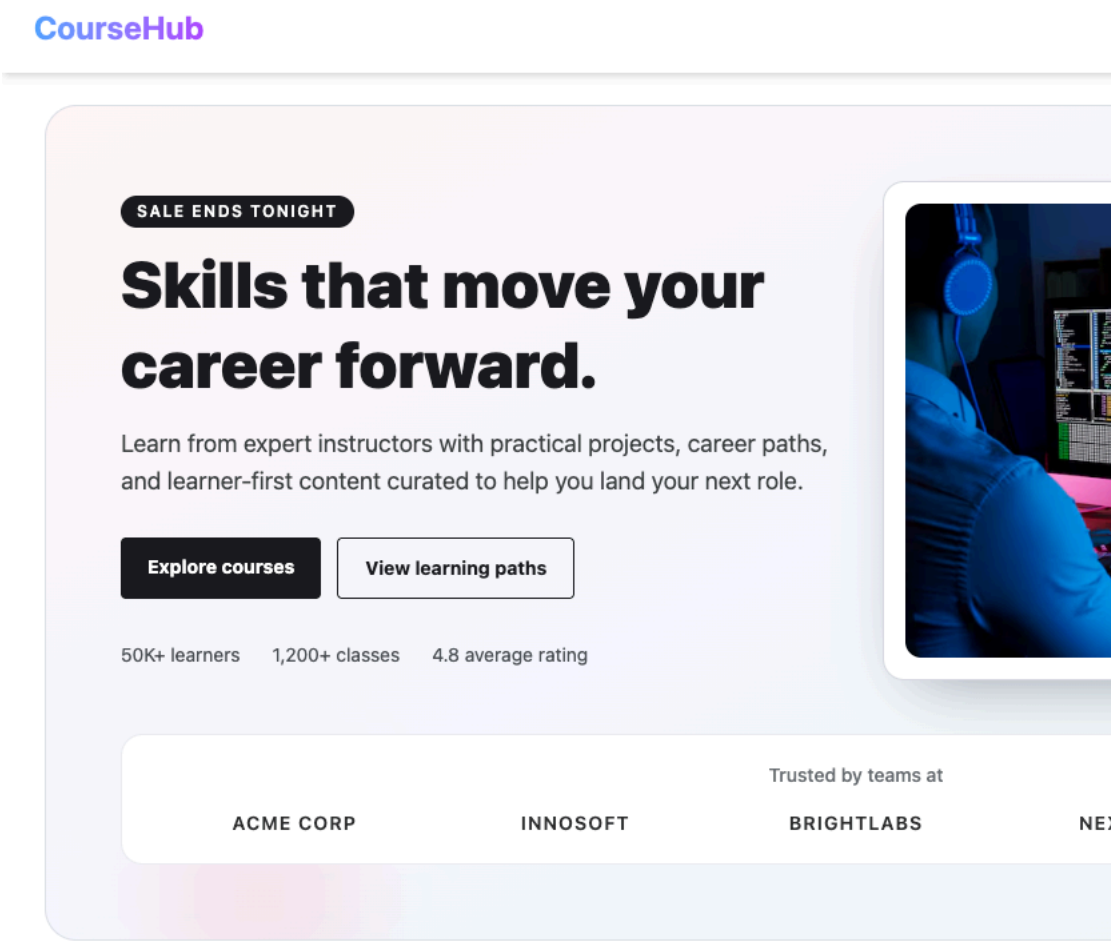


Figure 6.1: Home Landing Page Interface

- **Key Fields & Components:**
  - **Dynamic Top Navigation Bar:** Routing links (`/all-courses`, `/learning-paths`, `/login`, `/register`, `/register-startup`).
  - **Hero Call-to-Action Group:** "Explore courses" and "View learning paths" primary buttons.

- **Social Proof Counter:** Displays "50K+ learners", "1,200+ classes", and "4.8 average rating".
- **Floating AI Mentor Trigger:** Fixed position bottom-right button providing instant academic tutoring via Google Gemini.

6.2 Screen 2: User Registration with Startup Portal Callout

The standard registration page (`/register`) caters to students, instructors, and platform administrators. To prevent confusion, it features an eye-catching emerald callout directing corporate founders to the dedicated startup onboarding workflow.

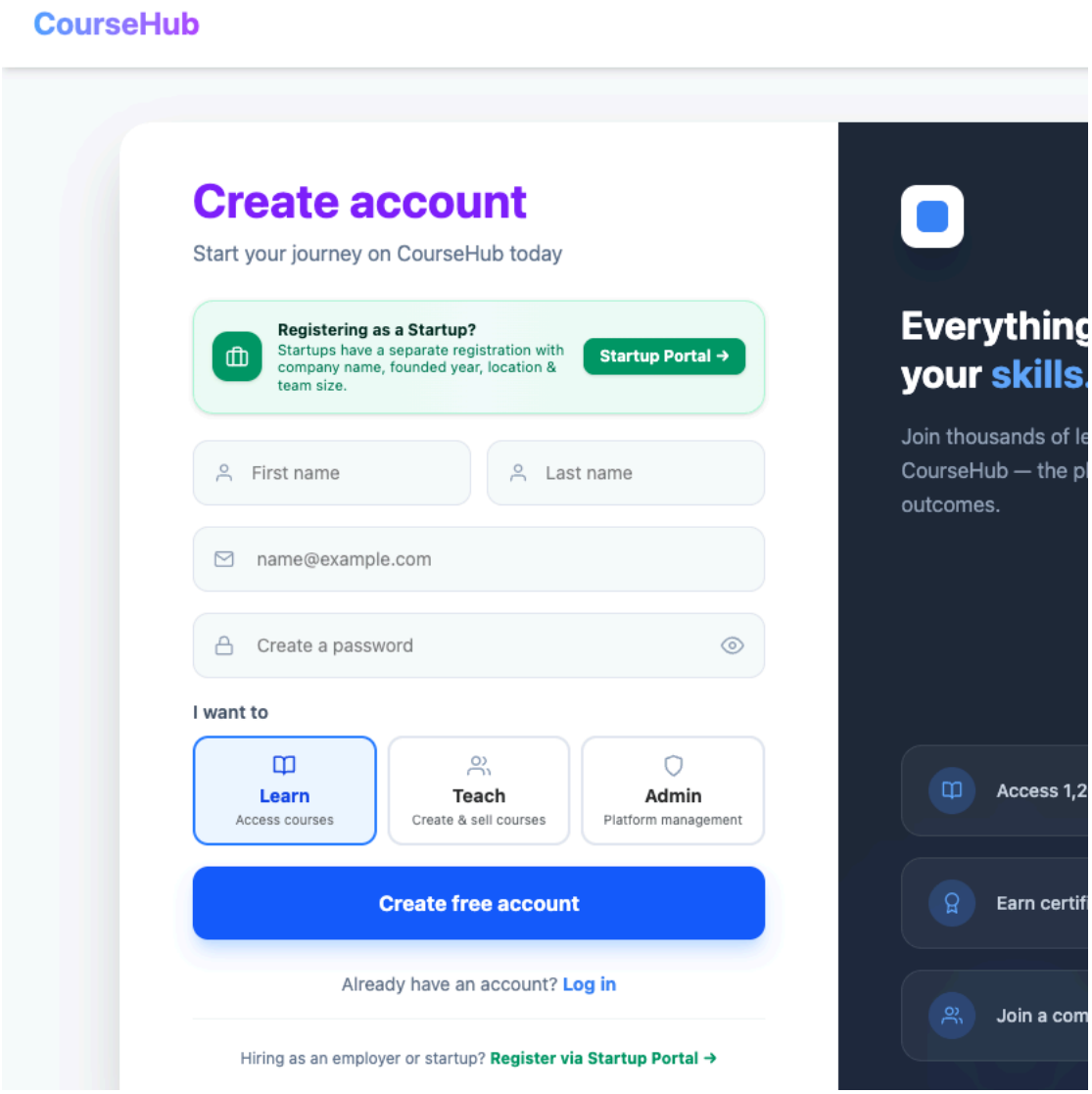


Figure 6.2: User Registration with Startup Portal Callout

- **Key Fields & Components:**
  - **Startup Gateway Banner:** "Registering as a Startup? Startups have a separate registration with company name, founded year, location & team size." with a direct button leading to `/register-startup`.
  - **Name Inputs:** First Name and Last Name inputs with iconography.
  - **Email & Password Fields:** Interactive password strength meter evaluating uppercase and numeric complexity.
  - **Role Selection Radio Cards:** Learn (Student), Teach (Instructor), Admin.

6.3 Screen 3: Dedicated Startup Registration Portal

Addressing the specific requirement for segregated corporate onboarding, this dedicated interface (`/register-startup`) collects organizational identity parameters required to vet hiring entities before they post paid opportunities.

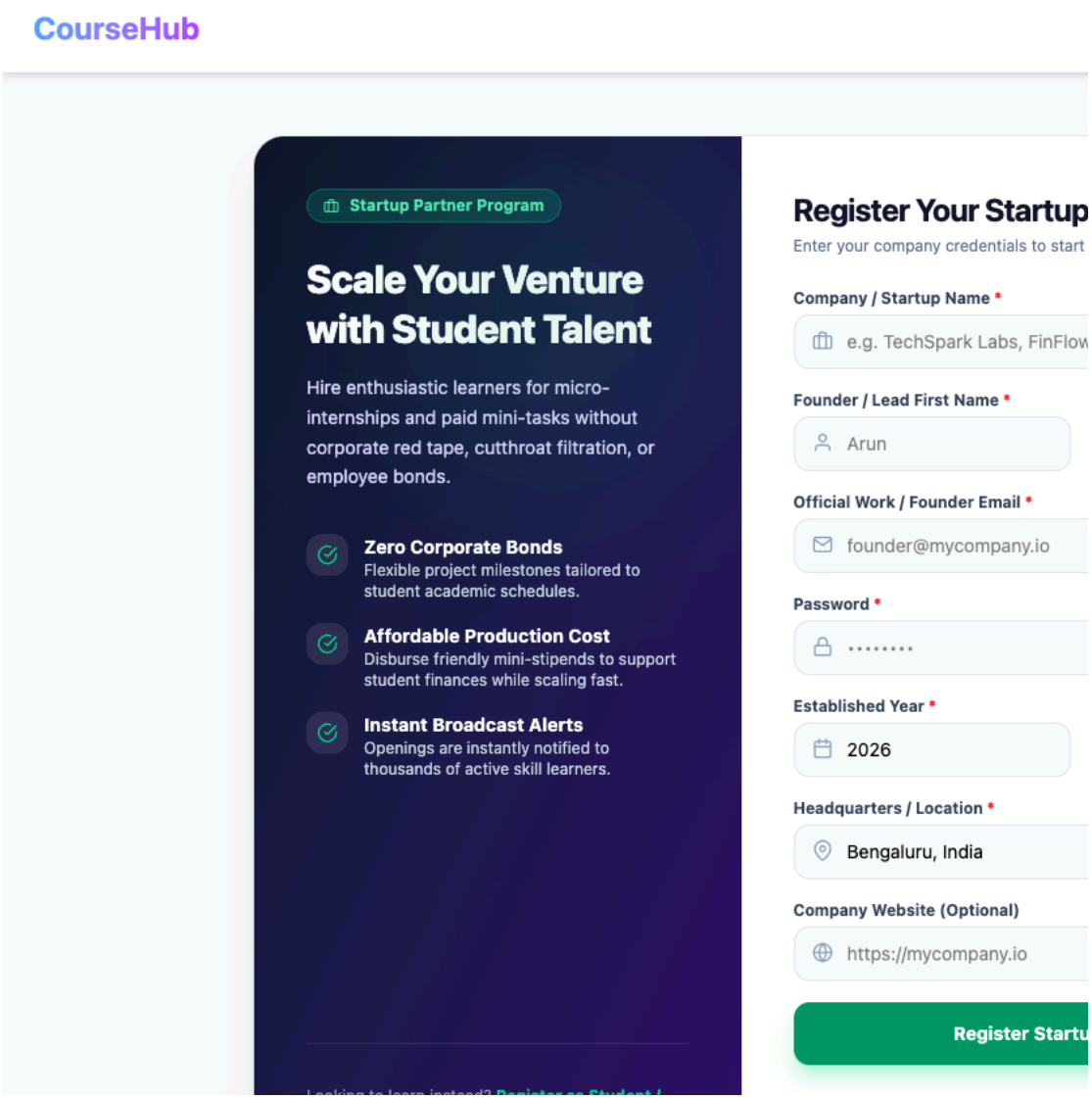


Figure 6.3: Dedicated Startup Registration Portal

- **Key Fields & Components:**
  - **Left Value Proposition Panel:** Highlights core student benefits: *Zero Corporate Bonds*, *Affordable Production Cost*, and *Instant Broadcast Alerts*.
  - **Company / Startup Name Input:** Captures legal business entity name (e.g., *vip, TechSpark Labs*).
  - **Founder / Lead First & Last Name:** Personal identity fields for the account administrator.
  - **Official Work Email:** Corporate email address enforced by regular expression validation.
  - **Password Input:** Secure field with minimum 6-character constraint and show/hide password toggle.
  - **Established Year Input:** Numerical year field restricted between 1950 and the current calendar year.
  - **Number of Employees Dropdown:** Segregates startup stage (`1-10 Early Stage`, `11-50 Seed / Growth`, `51-200 Expansion`, `200+ Scaleup`).
  - **Headquarters / Location Input:** Captures corporate base (e.g., *Bengaluru, India or Remote*).
  - **Submission Action:** "Register Startup Account →" button dispatching `POST /common-api/register-startup`.

6.4 Screen 4: User Authentication & Sign-In Interface

The login interface (`/login`) provides secure access across all four platform roles. After verifying credentials against bcrypt hashes, the backend returns a signed JWT and user metadata, redirecting users according to their respective role.

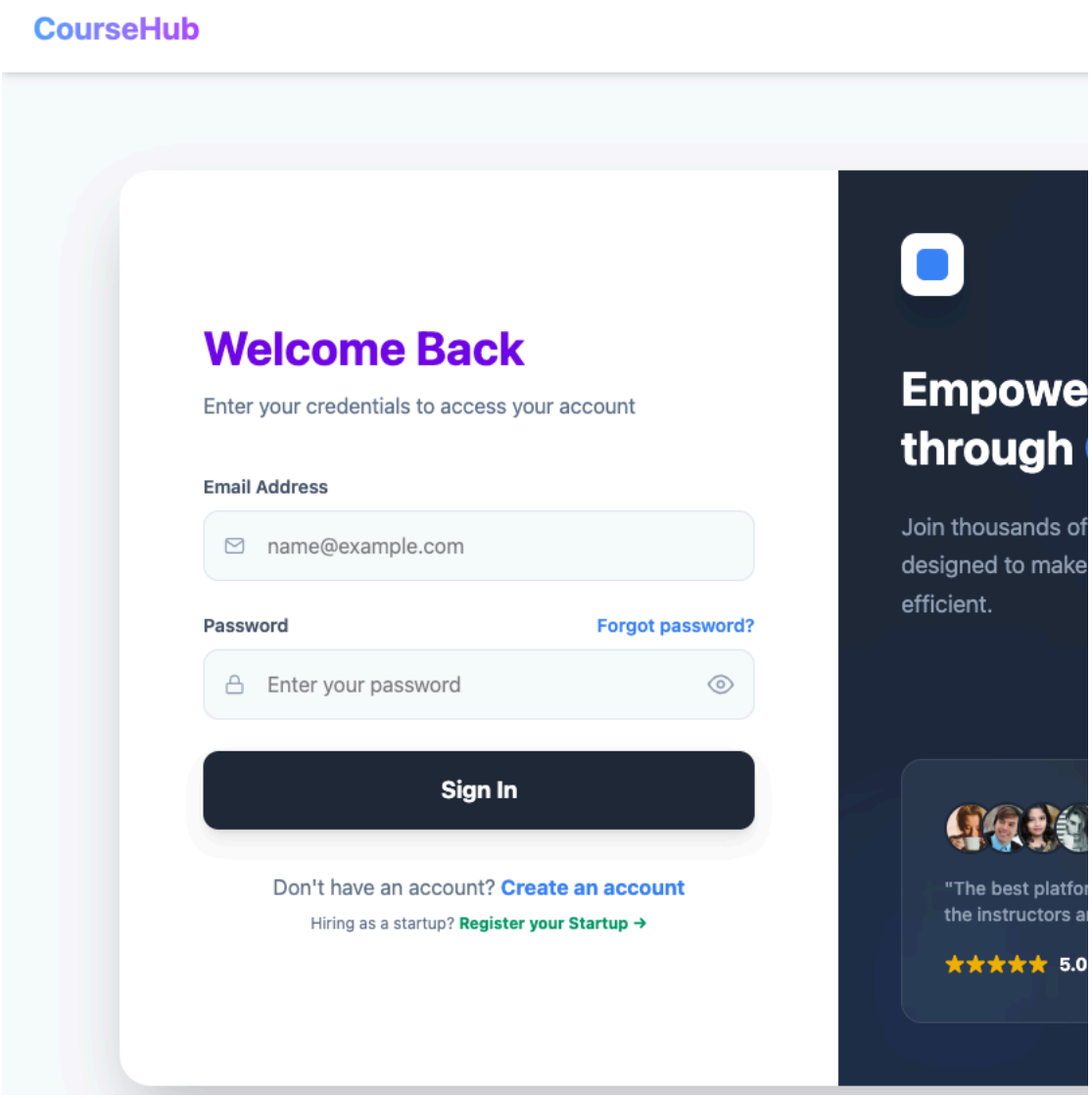


Figure 6.4: User Authentication & Sign-In Interface

- **Key Fields & Components:**
  - **Email Address & Password Fields:** Auto-focus inputs with validation and error feedback.
  - **Sign In Button:** Features asynchronous progress feedback and prevents duplicate submissions.
  - **Alternative Navigation:** Direct link for startups: *"Hiring as a startup? Register your Startup →"*.
  - **Testimonial Showcase:** Highlights verified learner community ratings.

6.5 Screen 5: Student Dashboard & Academic Progress Center

Upon logging in with student credentials (`rk@gmail.com`), learners are greeted with a personalized learning command center displaying current enrolled course metrics, streak indicators, and continue-learning modules.

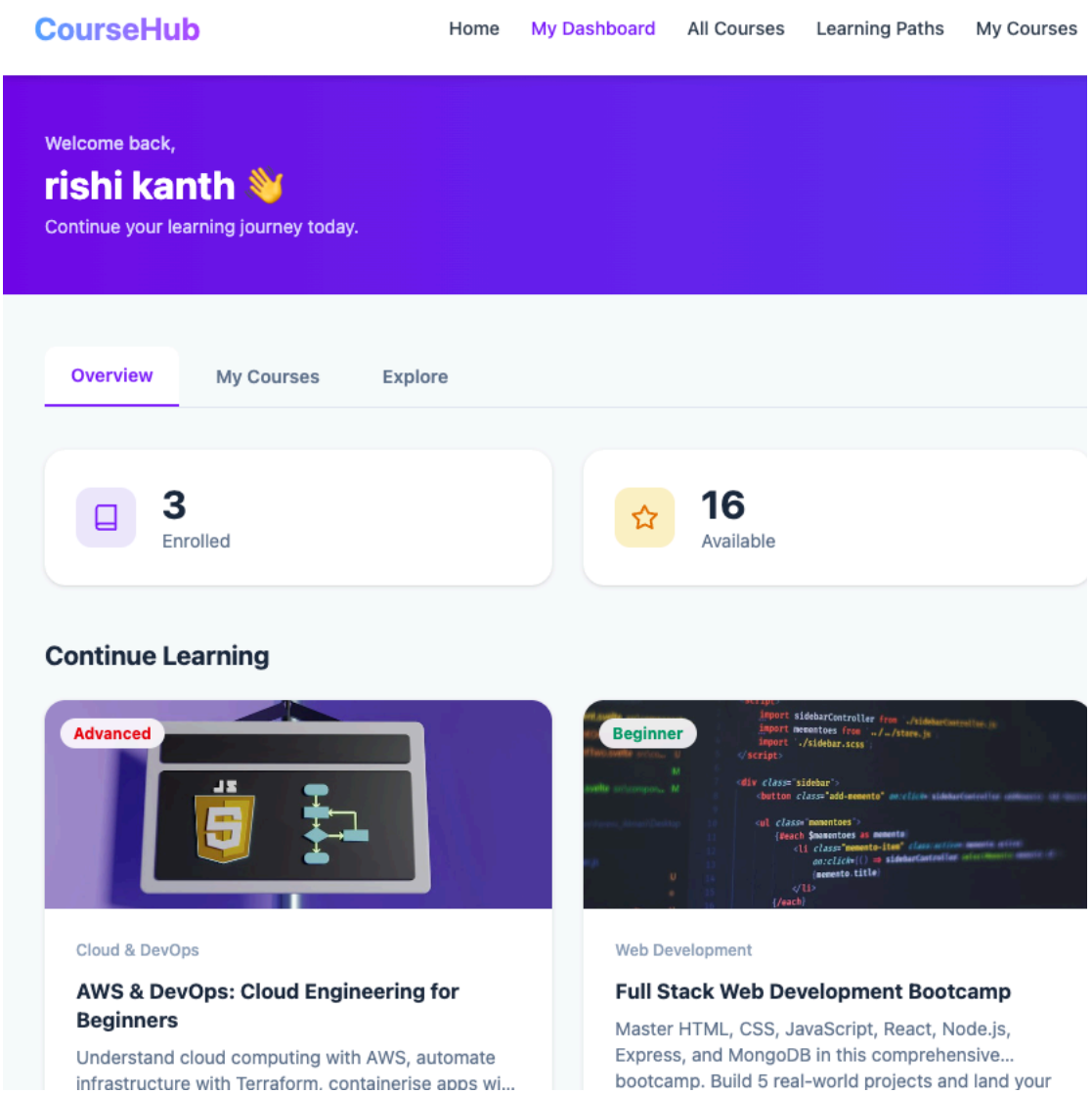


Figure 6.5: Student Dashboard & Progress Center

- **Key Fields & Components:**
  - **Personalized Header Banner:** "Welcome back, rishi kanth 👋 " with dynamic day-streak badge and enrolled course summary counters.
  - **Quick Metric Cards:** Enrolled courses count (3), Available catalog courses (16), and Active in-progress modules (3).
  - **Continue Learning Carousel:** Interactive cards for enrolled courses with direct resume playback triggers.
  - **AI Chatbot Bubble:** Bottom-right floating icon launching instant tutoring sessions.

6.6 Screen 6: Student Enrolled Courses & Study Portal

Accessible via `/my-courses`, this interface displays all curricular modules currently enrolled by the authenticated student.

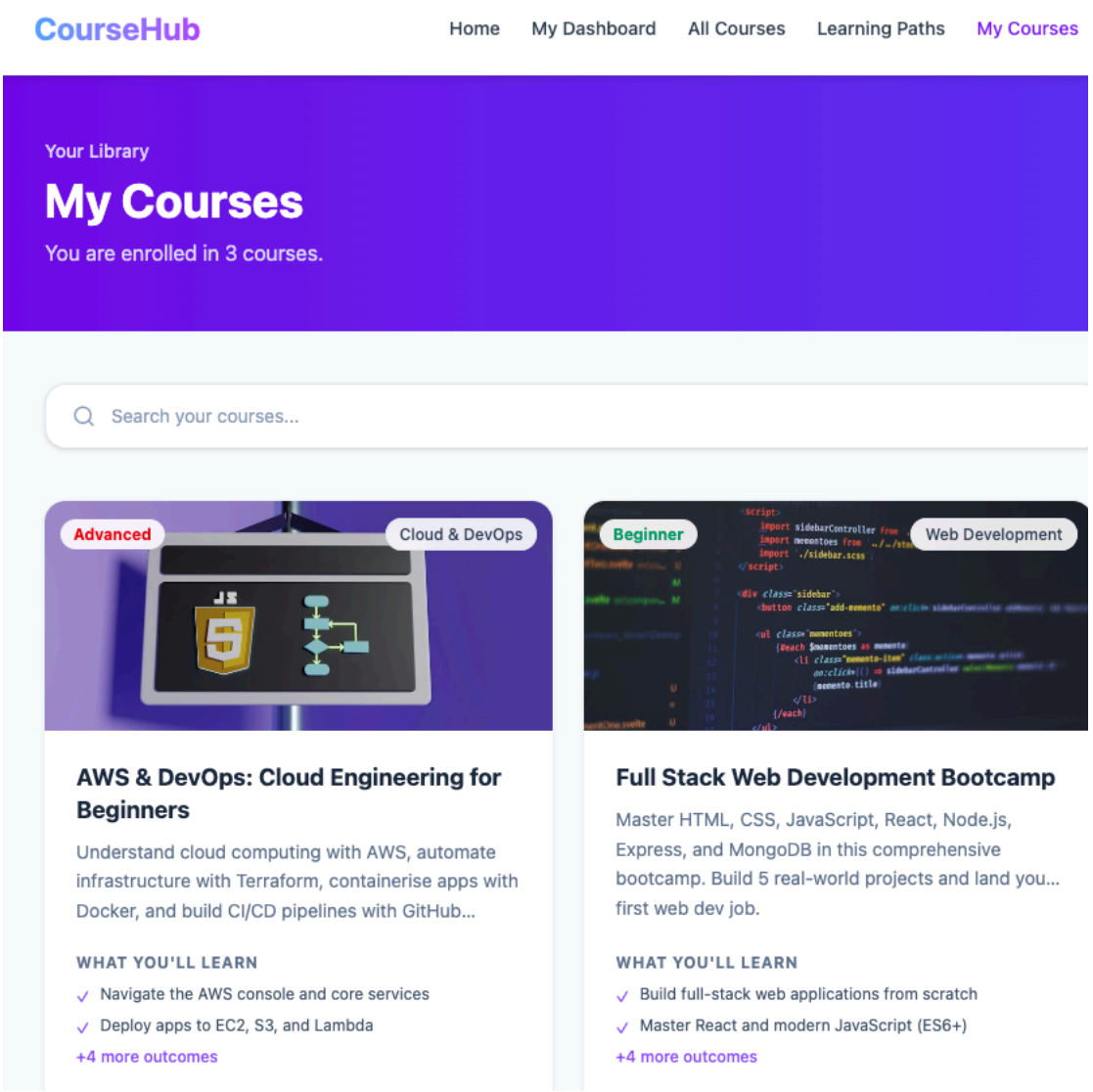


Figure 6.6: Student Enrolled Courses & Study Portal

- **Key Fields & Components:**
  - **Course Progress Gauges:** Percentage completion bars for individual lecture modules.
  - **Course Action Buttons:** "Continue Learning", "Course Details", and "Take Assessment".
  - **Search & Filter Controls:** Allows students to filter courses by subject, completion status, or instructor.

6.7 Screen 7: Comprehensive Course Catalog & Filtering

The course catalog (`/all-courses`) displays all published educational programs across cloud computing, machine learning, web engineering, and data science.

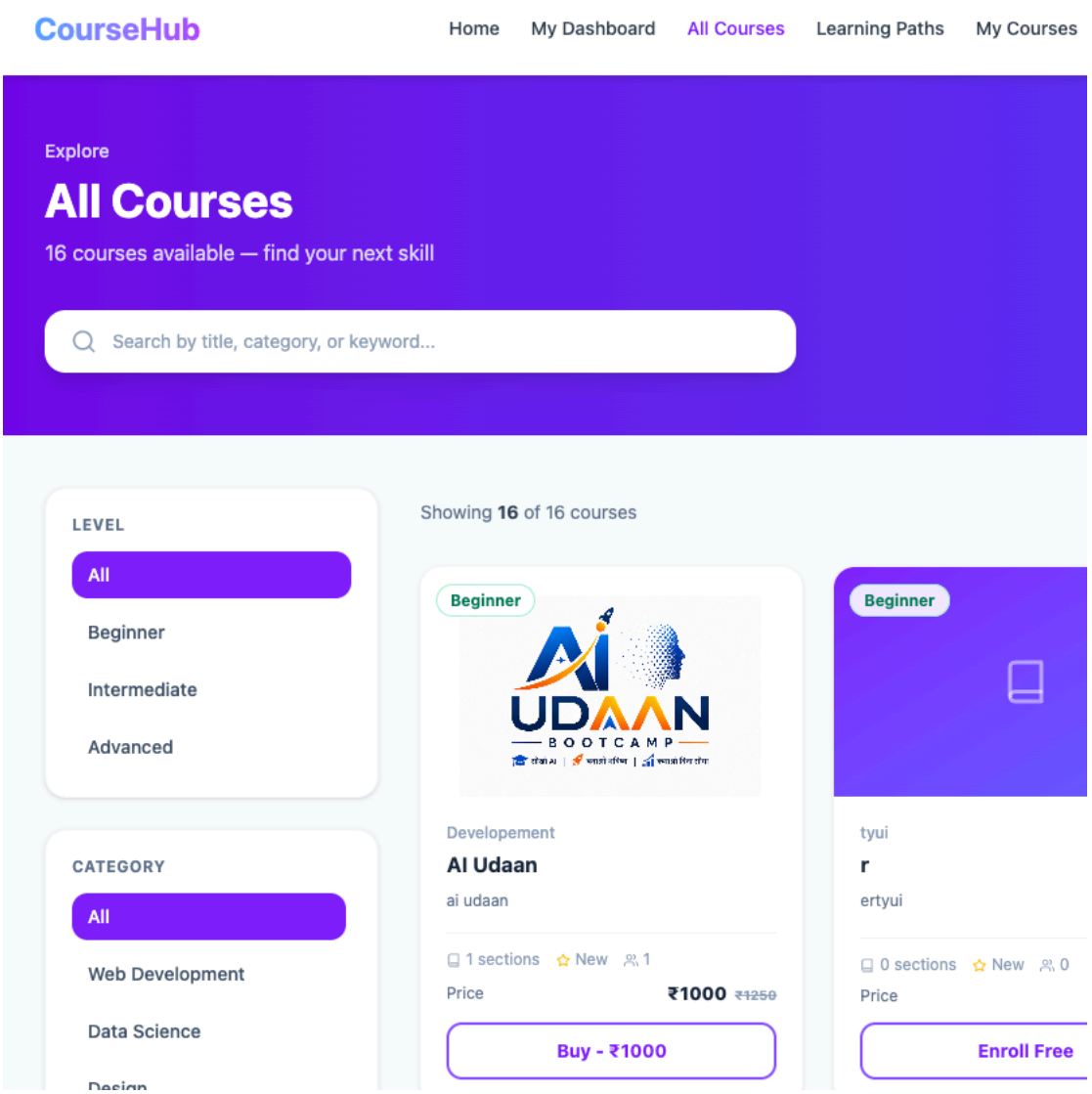


Figure 6.7: Comprehensive Course Catalog & Filtering

- **Key Fields & Components:**
  - **Search Bar:** Real-time query matching against titles, tags, and instructor names.
  - **Category Pills:** Rapid selection across technical domains (Full-Stack, Cloud, AI/ML, DevOps).
  - **Pricing & Rating Badges:** Transparent pricing, duration hours, and star review scores.

6.8 Screen 8: Structured Career & Learning Paths

The `/learning-paths` screen guides students through curated multi-course career roadmaps designed to take learners from beginner fundamentals to industry-ready engineering proficiency.

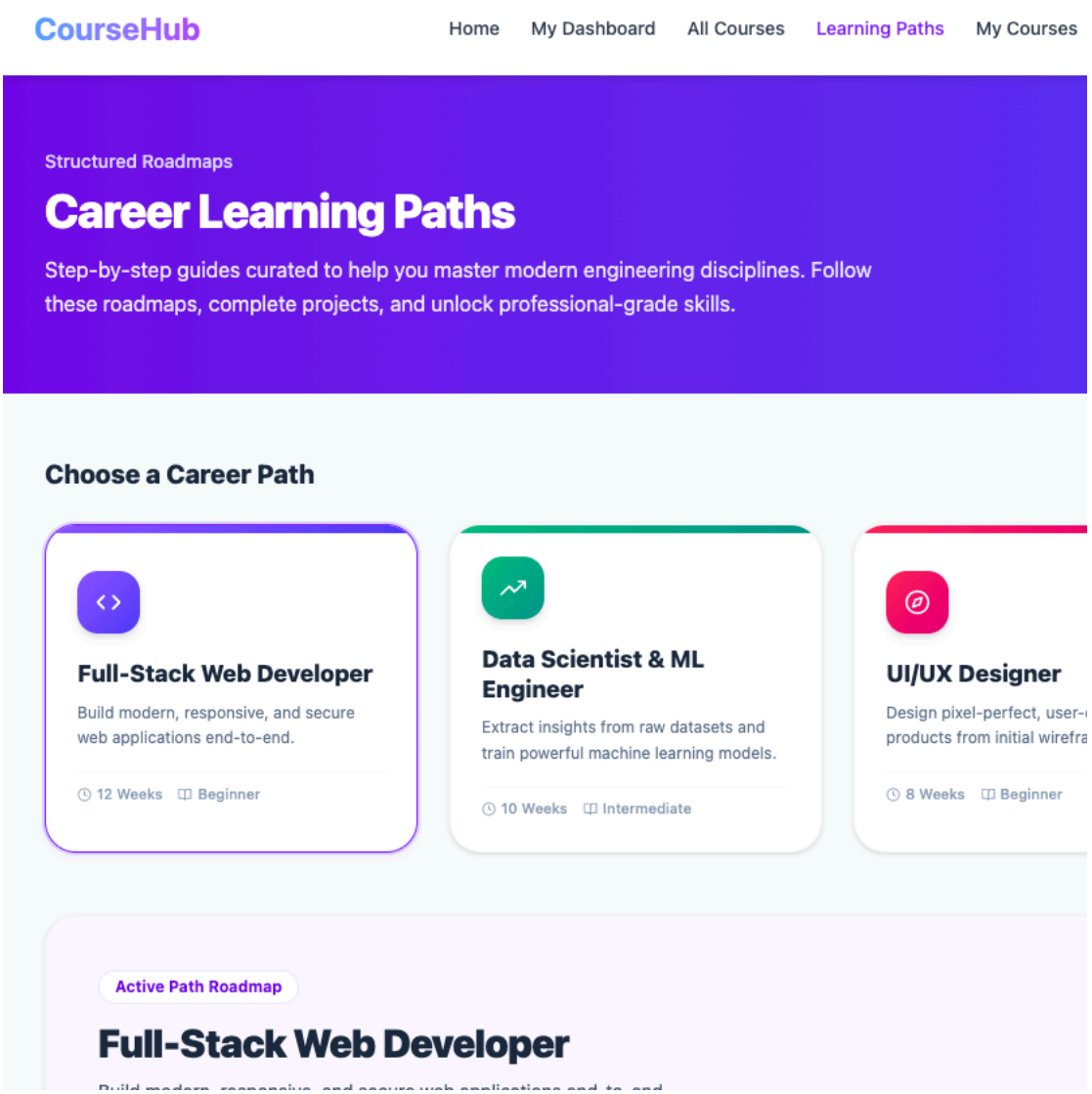


Figure 6.8: Structured Career & Learning Paths

- **Key Fields & Components:**
  - **Track Roadmap Visualizer:** Step-by-step milestones (e.g., Frontend Specialist, Cloud Architect, AI Engineer).
  - **Target Roles & Salary Ranges:** Estimated industry compensation and prerequisites.
  - **Enroll in Path Trigger:** One-click enrollment into grouped curricular bundles.

6.9 Screen 9: Startup Micro-Internships & Gig Marketplace

The `/jobs` portal represents the platform's core economic innovation: a "Skill-to-Earn" marketplace connecting verified learners directly to emerging startups for paid mini-tasks and micro-internships without corporate MNC filtration or bonds.

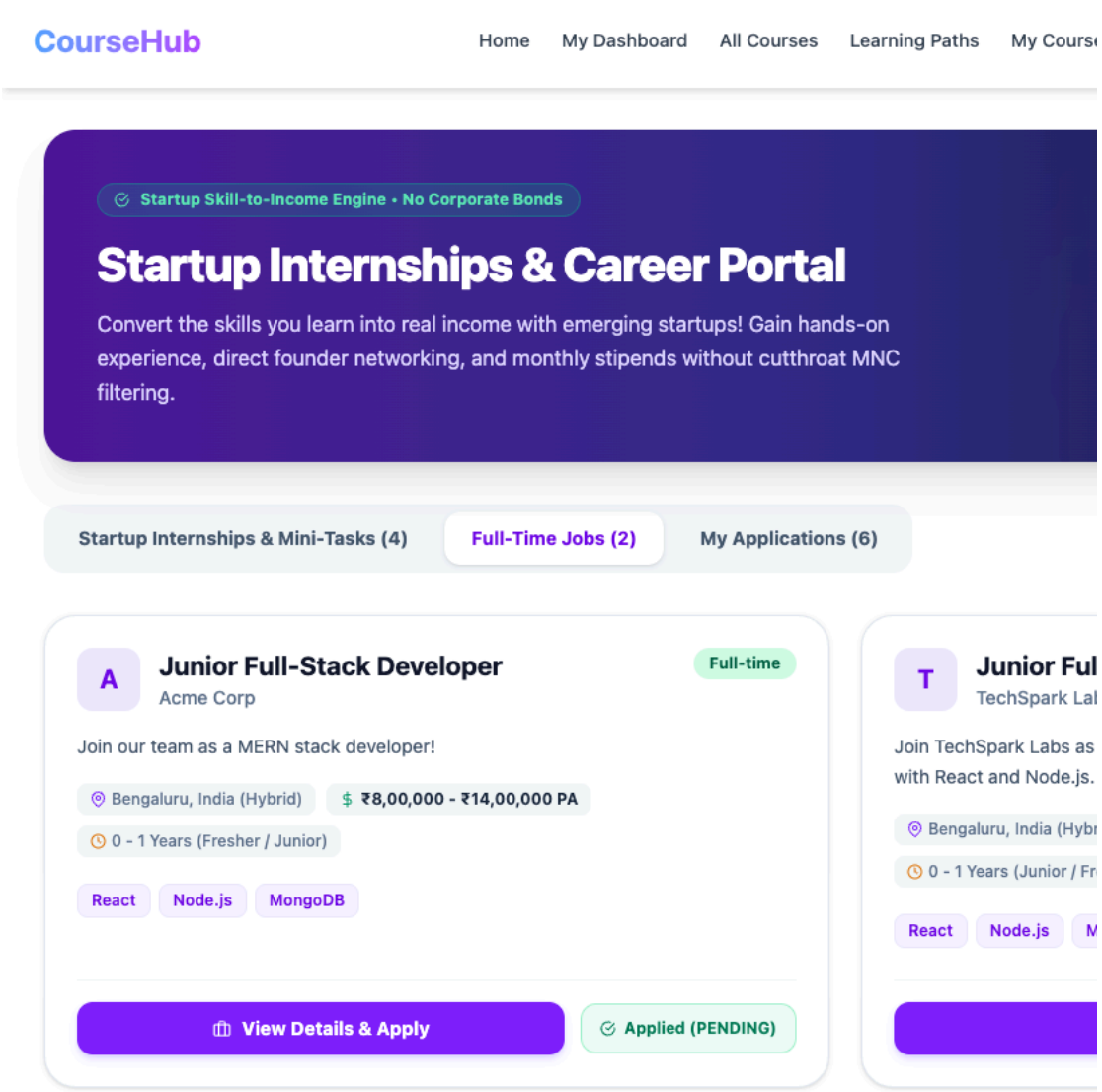


Figure 6.9: Startup Micro-Internships & Gig Marketplace

- **Key Fields & Components:**
  - **Category Tabs:** Filter between *Startup Internships & Mini-Tasks*, *Full-Time Jobs*, and *My Applications*.
  - **Opportunity Cards:** Displays company badge, job title, location mode (Remote/Hybrid), stipend/compensation, and required tech stack tags.
  - **Application Status Indicator:** Live status badges (e.g., `Applied (PENDING)` or `ACCEPTED`).
  - **View Details & Apply Button:** Opens detailed scope description and direct submission modal.

6.10 Screen 10: Startup Founder Portal & Applicant Management Hub

Authenticated startup founders (`vip@gmail.com`) manage their recruitment pipeline through `/startup-dashboard`. Founders can post new opportunities, review applicants, evaluate notes, and accept candidates.

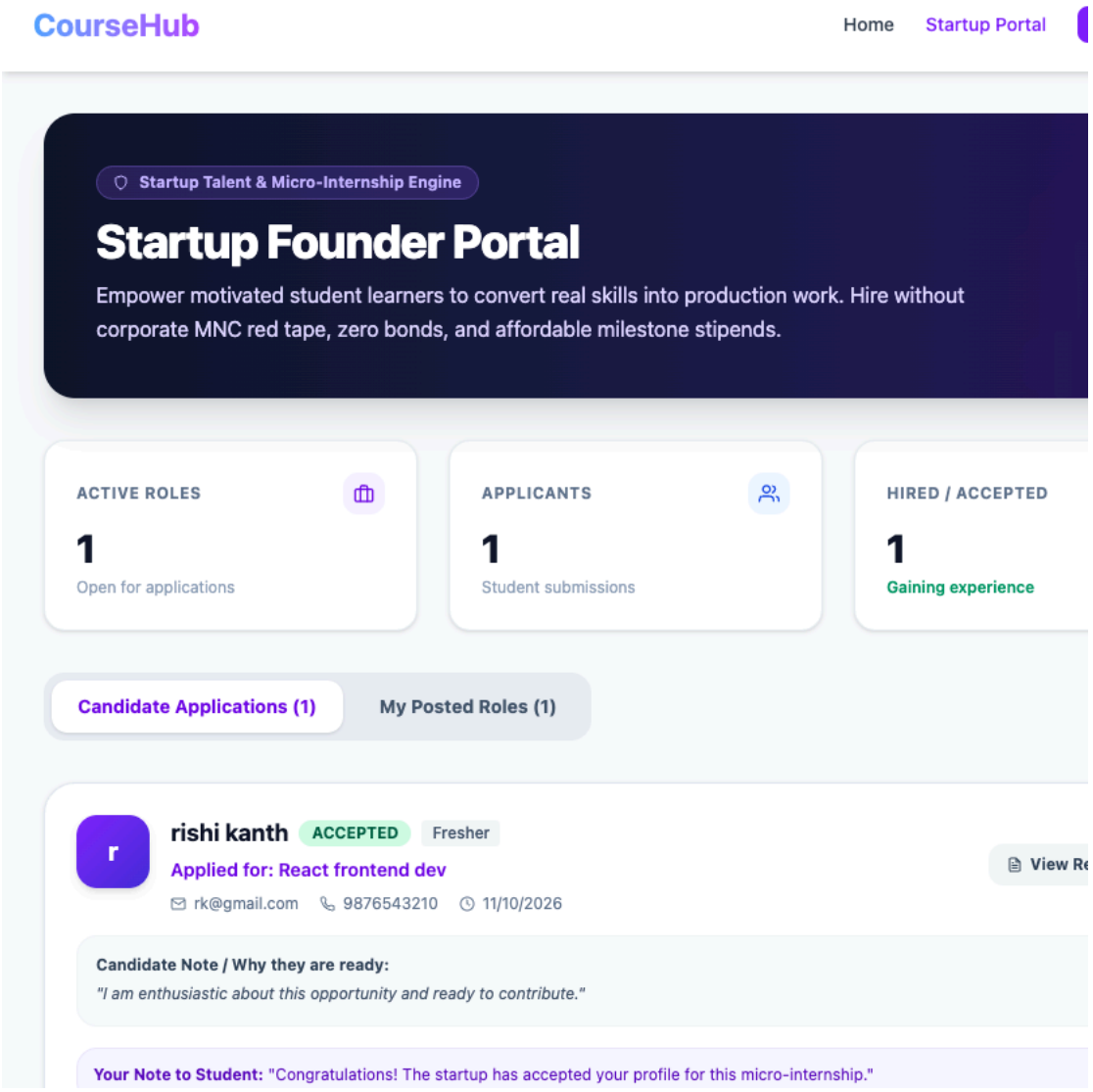


Figure 6.10: Startup Founder Portal & Applicant Management Hub

- **Key Fields & Components:**
  - **KPI Metric Dashboard:** Real-time counters for Active Roles, Total Applicants, Hired / Accepted candidates, and Under Review.
  - **Candidate Applications List:** Candidate cards showing student name, contact info, applied role, and current hiring status (e.g., student `rishi kanth` marked `ACCEPTED`).
  - **Hiring Action Buttons:** Quick action triggers: "Accept & Hire", "Shortlist", and "Note".
  - **Primary CTA:** "+ Post New Internship / Mini-Task" button in header and banner.

6.11 Screen 11: Startup Micro-Internship Opportunity Builder

Located at `/startup/post-internship`, this form allows founders to author and broadcast paid opportunities to learners within minutes.

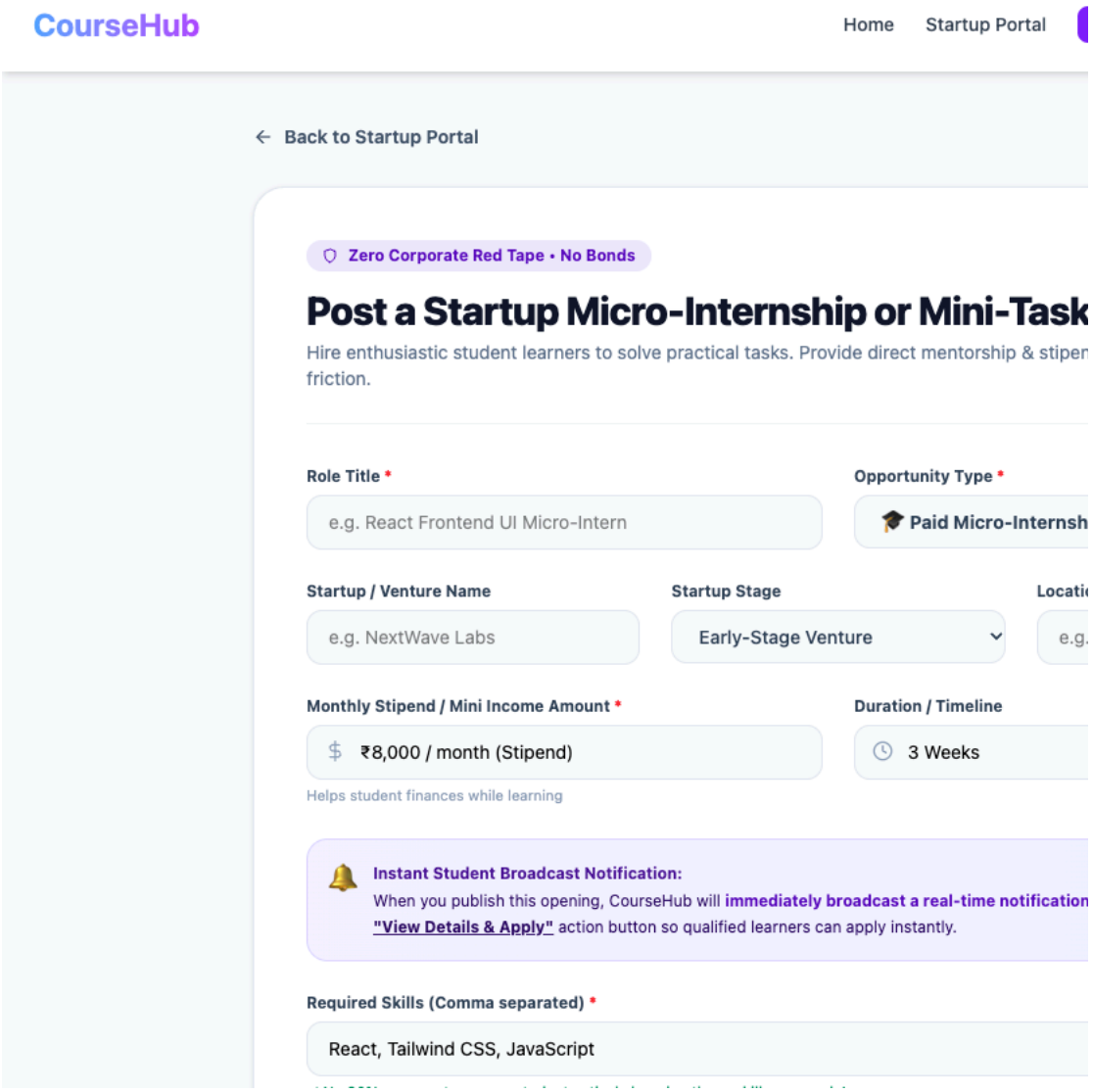


Figure 6.11: Startup Micro-Internship Opportunity Builder

- **Key Fields & Components:**
  - **Role Title & Domain:** Text input capturing specific role title (e.g., React Developer, AI Integrator).
  - **Compensation / Stipend Range:** Stipend fields specifying monthly or milestone payout.
  - **Work Mode & Duration:** Select dropdowns for Remote, On-Site, Hybrid, and duration weeks.
  - **Required Skills & Scope Description:** Rich text area detailing project deliverables.

6.12 Screen 12: Instructor Dashboard & Course Analytics

Educators logging in via instructor credentials (`pr@gmail.com`) access `/instructor-dashboard`, providing revenue analytics, student enrollment tallies, and course performance reviews.

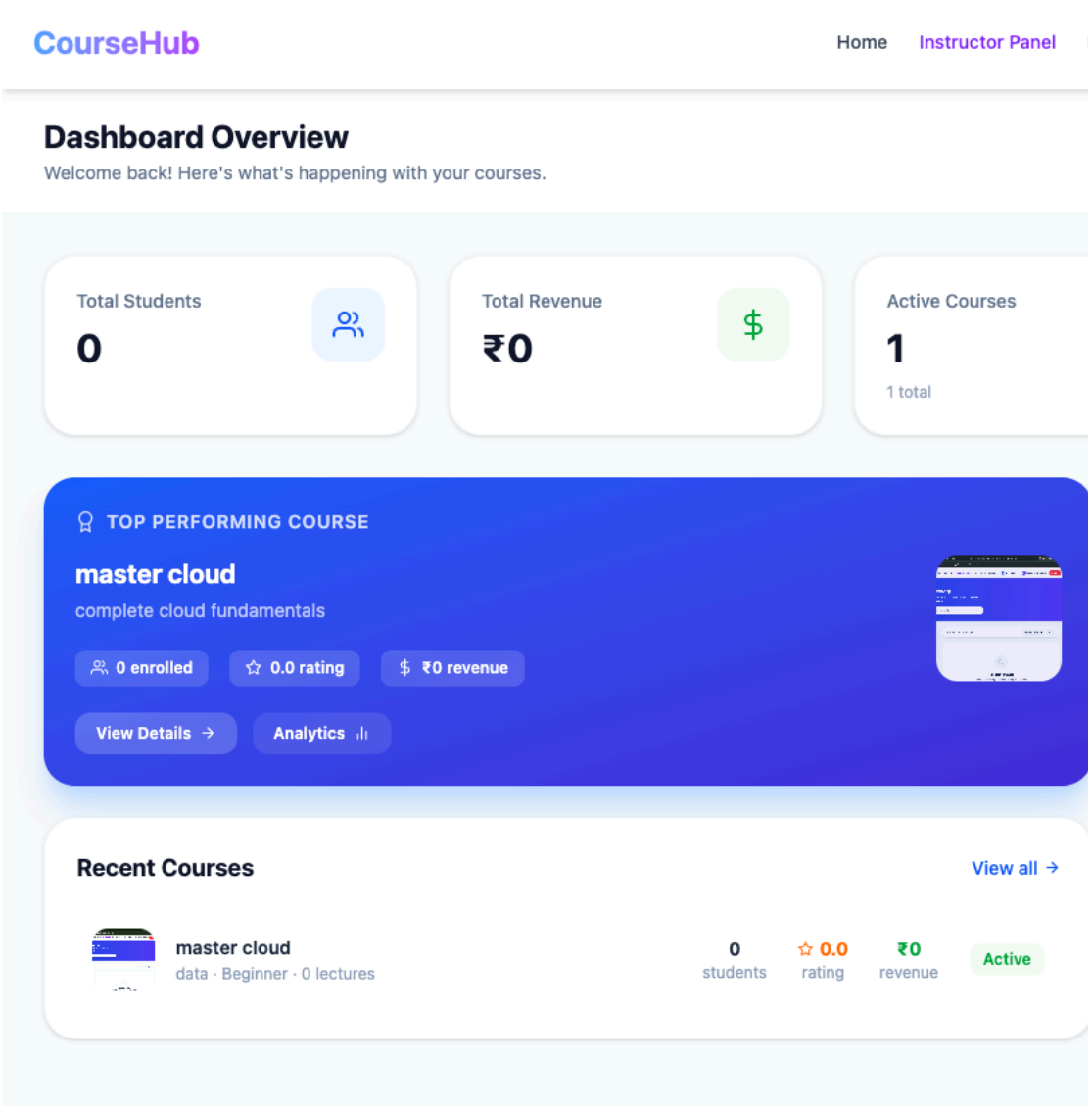


Figure 6.12: Instructor Dashboard & Course Analytics

- **Key Fields & Components:**
  - **KPI Metric Strip:** Total Students, Total Revenue (₹), Active Courses, and Average Rating.
  - **Top Performing Course Spotlight:** Featured card showing enrolled metrics and analytics link.
  - **Quick Action Bar:** Fast shortcuts to "+ Create New Course", "Manage My Courses", and "Edit Profile".

6.13 Screen 13: Instructor Curriculum & Course Creation Interface

At `/create-course`, instructors construct multimedia learning programs, upload video lectures to Cloudinary, design curricula, and set assessment parameters.

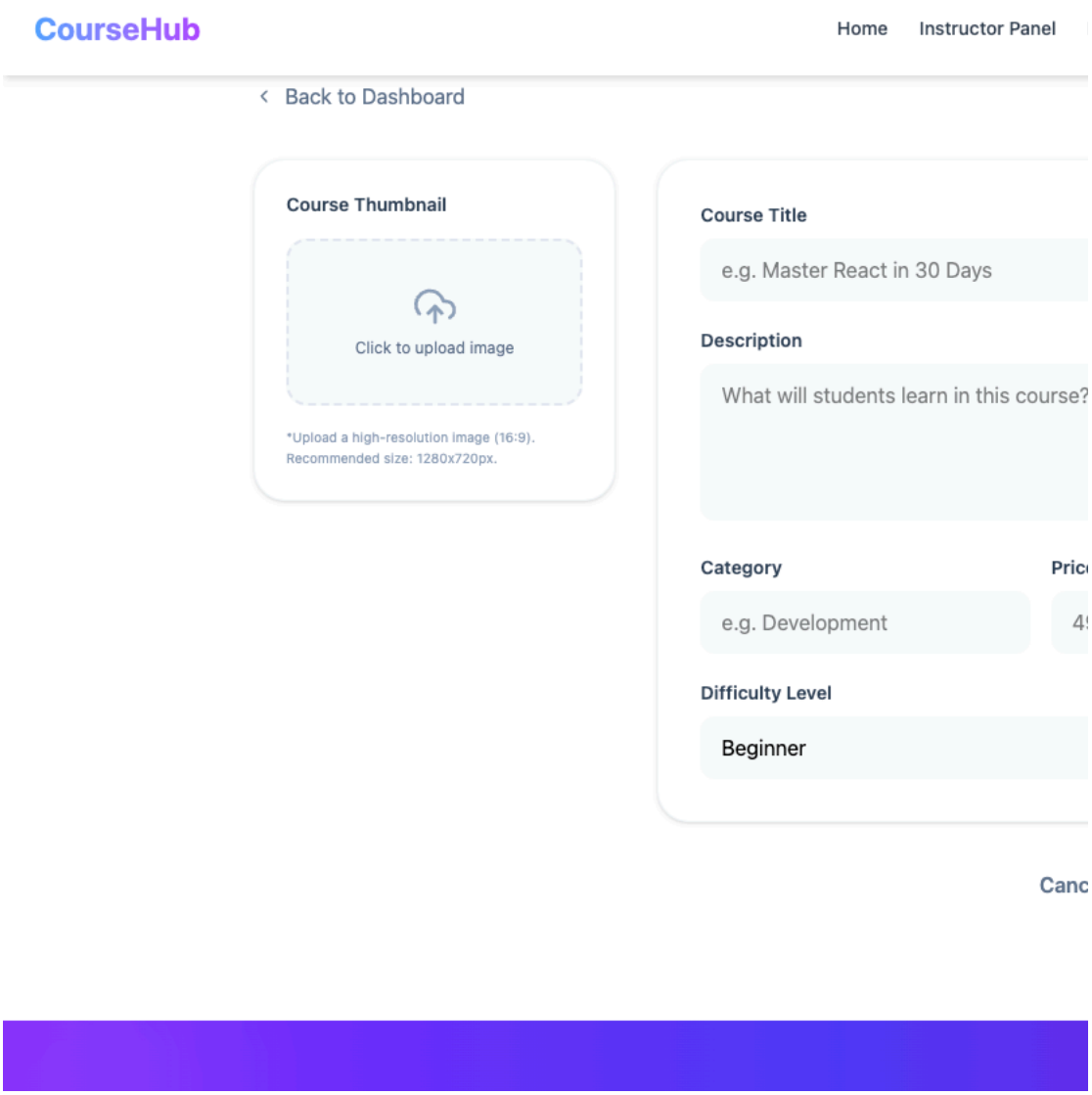


Figure 6.13: Instructor Curriculum & Course Creation Interface

- **Key Fields & Components:**
  - **Course Metadata Editor:** Course title, category, target difficulty level, and description.
  - **Pricing & Publishing Controls:** Free vs Paid toggle and course fee inputs.
  - **Section & Lesson Builder:** Drag-and-drop modular syllabus organization.

6.14 Screen 14: Instructor Course Management & Publishing Center

The `/instructor-my-courses` page allows educators to manage their published educational catalog, inspect individual student completion rates, and trigger content updates.

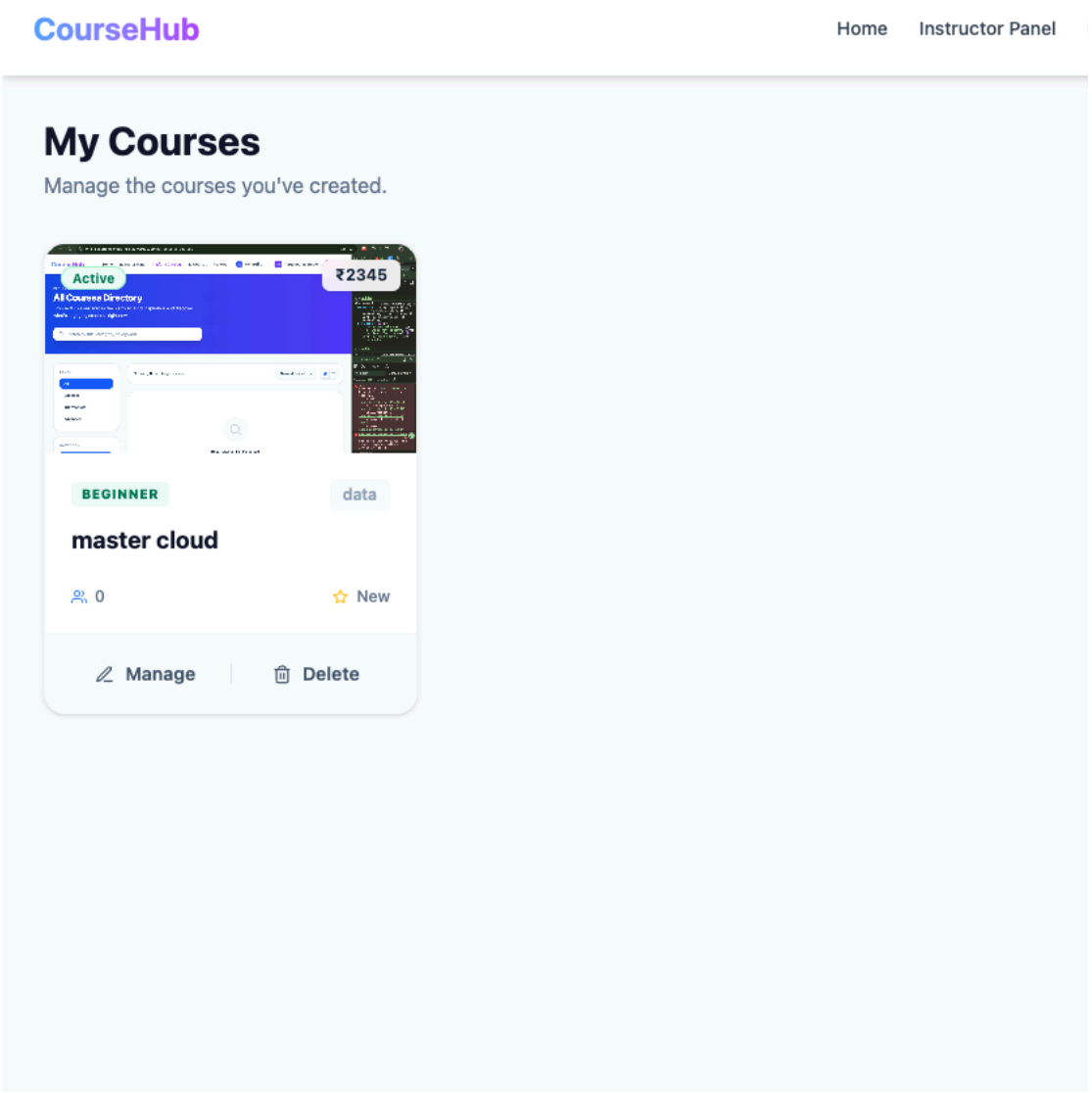


Figure 6.14: Instructor Course Management & Publishing Center

- **Key Fields & Components:**
  - **Course Inventory Table:** Tabular listing of active, draft, and archived courses.
  - **Status Pills:** Live status indicators reflecting moderation and availability state.
  - **Management Actions:** Edit syllabus, view analytics, and update pricing.

6.15 Screen 15: Cryptographic Certificate Verification Portal

CourseHub provides a tamper-proof verification portal (`/verify-certificate`) enabling recruiters, academic institutions, and employers to confirm credential legitimacy using certificate hash IDs or QR codes.

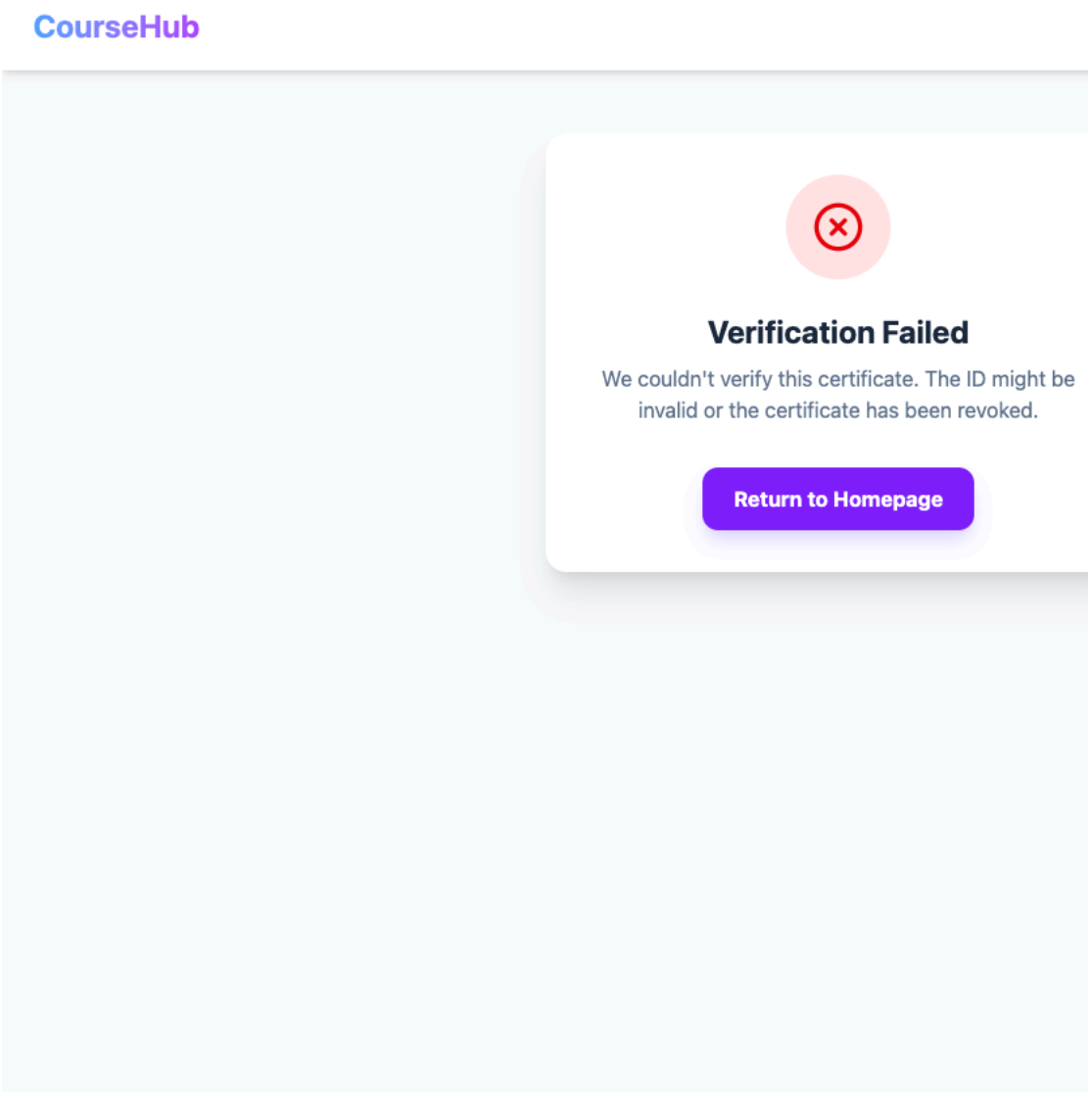


Figure 6.15: Cryptographic Certificate Verification Portal

- **Key Fields & Components:**
  - **Certificate ID Search Field:** Allows manual entry of alphanumeric verification hash.
  - **Verify Certificate Button:** Dispatches validation request to `/common-api/verify-certificate/:id`.
  - **Verification Card:** Displays student name, course title, issuance date, and official seal upon successful validation.

CHAPTER 7

TESTING METHODOLOGY

7.1 UNIT TESTING

Unit tests were executed on discrete backend controller functions and utility routines.

- **Password Encryption:** Verified that `bcrypt.hash()` reliably hashes plain-text inputs with salt factor 10.
- **JWT Verification:** Validated that expired or tampered tokens return HTTP 401 Unauthorized.
- **Input Sanitization:** Ensured malformed email patterns return HTTP 400 Bad Request.

7.2 INTEGRATION TESTING

Integration testing verified communication between decoupled services.

- **Authentication to Database Handshake:** Tested `POST /common-api/register-startup` saving to MongoDB Atlas.
- **Client to Cloud Storage Pipeline:** Validated file uploads reaching Cloudinary and returning secure HTTPS asset URLs.
- **AI Tutoring Pipeline:** Tested student prompts reaching Google Gemini API and streaming responses back within 1.5 seconds.

7.3 ROLE-BASED SECURITY TESTING

Evaluated access boundaries to ensure unauthorized users cannot reach restricted operations:

- Non-authenticated requests to `/user-api/profile` correctly return 401.
- Students attempting to call `/instructor-api/create-course` receive 403 Forbidden.
- Standard students attempting to access `/startup-dashboard` are redirected to `/student-dashboard`.

7.4 TEST CASES AND EXECUTION RESULTS

Table 7.1: Comprehensive System Test Cases and Results

Test Case ID	Test Scenario	Input Data	Expected Result	Actual Result	Status
TC-01	Student Registration	Valid name, email, password	User created, status 201	User created in DB	**PASS**
TC-02	Duplicate Email Registration	Existing registered email	Error 400, "Account already exists"	Error 400 returned	**PASS**
TC-03	Startup Dedicated Registration	Company name, year, employees, location	User role 'STARTUP', startupProfile created	Role 'STARTUP' created	**PASS**
TC-04	Valid User Login	Correct email & password	Token issued, HTTP 200	Token returned, logged in	**PASS**
TC-05	Invalid Password Login	Correct email, wrong password	Error 401, "Invalid email or password"	Error 401 returned	**PASS**
TC-06	Course Enrollment	Valid `courseId`	Course ID appended to user enrollments	Enrollment saved in DB	**PASS**
TC-07	Video Progress Tracking	Chapter completion event	Progress updated to 100%	Progress updated in DB	**PASS**
TC-08	Post Startup Gig	Title, stipend, deliverables	Gig saved, student notifications broadcast	Gig created & broadcast	**PASS**
TC-09	Apply for Gig	Resume URL, GitHub link	Application saved as 'APPLIED'	Application record created	**PASS**

Test Case ID	Test Scenario	Input Data	Expected Result	Actual Result	Status
TC-10	Update Candidate Status	Status set to 'HIRED'	Application status updated, student notified	Status updated in DB	**PASS**
TC-11	QR Certificate Verification	Valid Certificate ID	Status 200, returns student & course metadata	Verified successfully	**PASS**
TC-12	Cloud Healthcheck	`GET /health`	Status 200, `{"status": "healthy"}`	200 OK returned	**PASS**

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1 CONCLUSION

The **Online Learning Platform with AI Mentorship and Startup Micro-Internships (CourseHub)** successfully resolves the disconnect between theoretical academic learning and commercial engineering practice. Built upon the modern **MERN Stack**, the platform demonstrates that web architectures can incorporate interactive video delivery, artificial intelligence tutoring, and freelance marketplace mechanics within a unified, high-performance ecosystem.

Key achievements realized in this project include:

1. **Practical "Skill-to-Earn" Pipeline:** Allowed students to undertake paid micro-internships for real startups, eliminating restrictive corporate bonds and 90% academic score gates.
2. **Dedicated Corporate Onboarding:** Engineered a dedicated startup registration workflow capturing critical business parameters (Company Name, Founded Year, Location, Team Size).
3. **AI-Powered Pedagogy:** Integrated Google Gemini to provide 24/7 personalized code tutoring and automated assessment generation.
4. **Verifiable Academic Integrity:** Introduced cryptographic QR-code certificates verifiable via public endpoints.
5. **Production Cloud Resilience:** Deployed the backend on **Render** and the frontend on **Vercel** with full cross-domain JWT security and high availability.

8.2 FUTURE ENHANCEMENTS

While the current implementation fulfills all core technical and academic objectives, future development will expand the platform's capabilities:

1. **Cross-Platform Mobile Application:** Develop native iOS and Android clients using React Native with background video caching for offline learning.
2. **Smart Contract Escrow Payments:** Implement automated milestone escrow releases using blockchain smart contracts or Razorpay/Stripe Route to instantly disburse student stipends upon verified GitHub pull request merges.
3. **WebRTC Live Classrooms:** Integrate peer-to-peer WebRTC video streaming and shared collaborative whiteboard tools for live instructor office hours.
4. **Browser-Based Cloud Coding Sandbox:** Embed a Monaco-powered online code execution sandbox (utilizing WebContainers) enabling students to solve code challenges directly in the browser without local software configuration.

CHAPTER 9

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